

Encapsulated bacteria theme overview

- Session 1: Germs to genes (Martin & Keith)
- Session 2: Introduction to Linux and BLAST (Keith & Martin)
- Session 3: Molecular approaches for detecting encapsulated bacteria (Dam, Brenda, Simon, Shola)
- Session 4: Interpretation of molecular tests for encapsulated bacteria (Dam, Brenda, Simon, Shola)
- Session 5: Introduction to Phylogenetics (Martin & Keith)
- Session 6: Genes to genomes, single gene to multilocus (Martin & Keith)
- Session 7: Multi locus to whole genome analysis (Martin & Keith)
- Session 8: Germs, Genes, and Genomes, translation and application to meningococcal disease in Africa (Martin& Keith)

Session 6: Single locus to multi-locus: 'Genes to Genomes'

Molecular Approaches to Clinical Microbiology in Africa 2025

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Learning Outcomes

After this lecture you should understand:

1. The nature and extent of bacterial diversity.
2. How sequence data illuminate bacterial diversity.
3. The variety of bacterial population structures and the implications of horizontal gene transfer (HGT).
4. Comparison of gene-by-gene and nucleotide by nucleotide analyses.
5. Practical implications of molecular epidemiology and population biology to clinical problems.

Impact of microbiology on clinical practice



- Why is the patient ill?
 - Germ theory enables the identification of the causative agent.
- What can we do about it?
 - Treatment pathways vary depending on the causative agent.
- What will happen next?
 - Knowledge of the agent and how it spreads informs public health measures, including prophylaxis.

Questions in clinical microbiology

evolution

emergence

epidemiology

diagnosis

Centuries+

decades

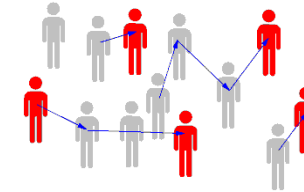
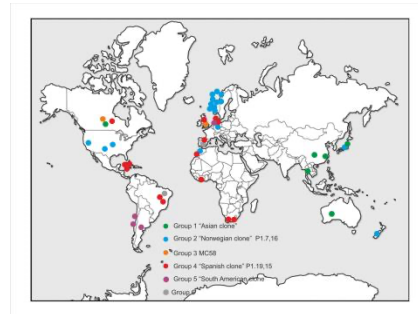
years

months

weeks

days

hours



High

Relative amount of genetic change

Low

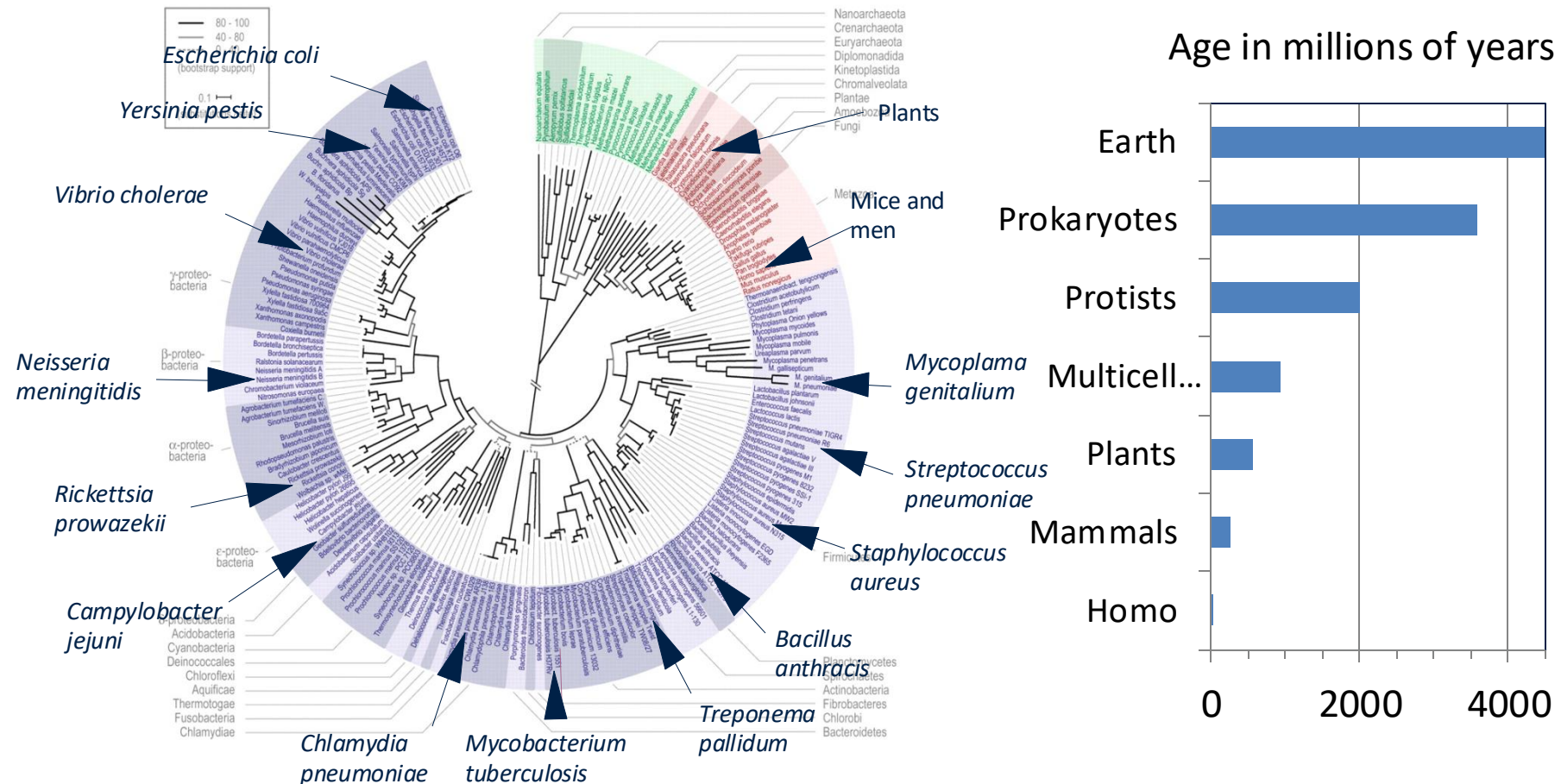
Low

Relative discrimination required

High



The scale of the problem: planet of the bacteria



Stephen Jay Gould, "Planet of the Bacteria," Washington Post Horizon, 1996, 119 (344)

Ciccarelli, F. D., Doerks, T., von Mering, C., Creevey, C. J., Snel, B. & Bork, P. (2006). Toward automatic reconstruction of a highly resolved tree of life. *Science* 311, 1283-1287.

The scale of the problem: bacterial genome diversity

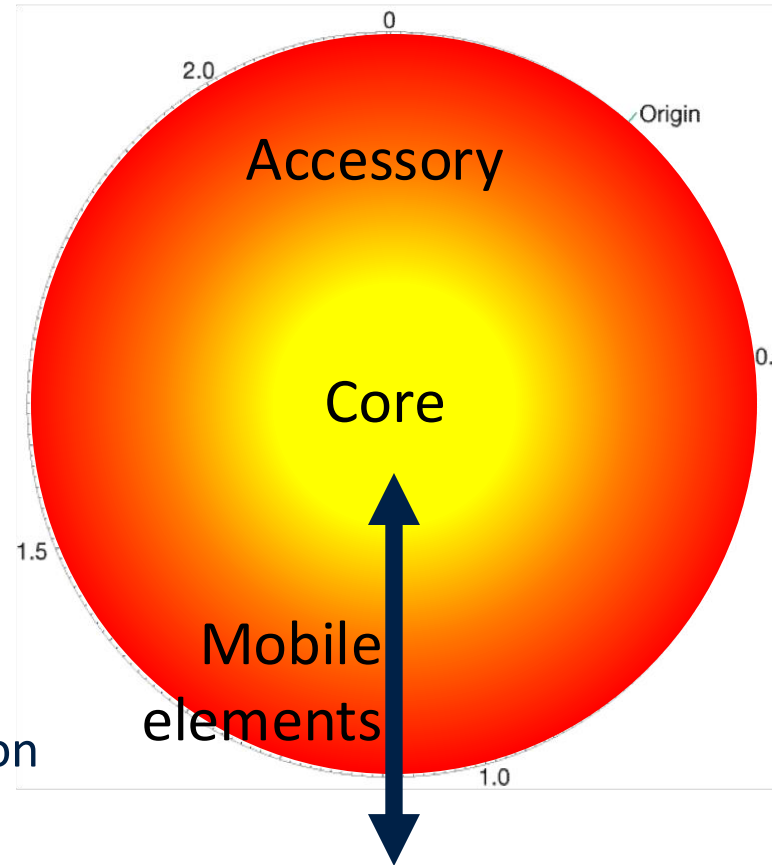
Core genome:

e.g. DNA replication, ribosomes, cell envelope, key metabolic pathways.

Parasitic elements (phages plasmids)

e.g. toxins, restriction/modification systems.

Gene pool



Accessory genome:

e.g. alternative metabolic pathways, transport systems.

Tettelin, H., Riley, D., Cattuto, C. & Medini, D. (2008). Comparative genomics: the bacterial pan-genome. *Current Opinion in Microbiology* 11, 472-477.

Gene pool:

e.g. antibiotic resistance, degradative metabolism.

Genetic elements may be subject to **stabilising (negative)** or **diversifying (positive)** selection or be **neutral** (rare in most bacteria).

Implications of bacterial diversity for clinical microbiology

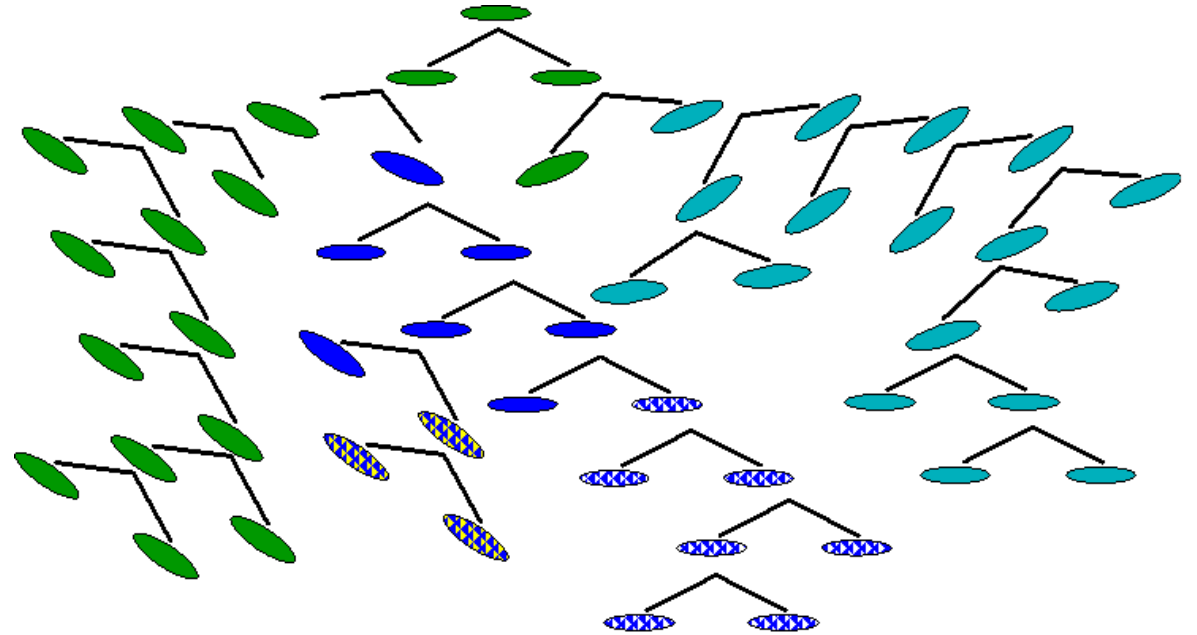
- The information required by practitioners includes:
 - what organism is causing disease;
 - what are its characteristics (antimicrobial resistance, vaccine antigens);
 - how is it related to other organisms (epidemiology, outbreak control).
- Determining this from sequence data depends on:
 - the degree of variation present;
 - how the variation is structured.
- This varies widely among bacteria, which determines:
 - what data are required;
 - how much data are required; and,
 - how they are analysed.

Patterns in sequence variation

a primer on bacterial population biology

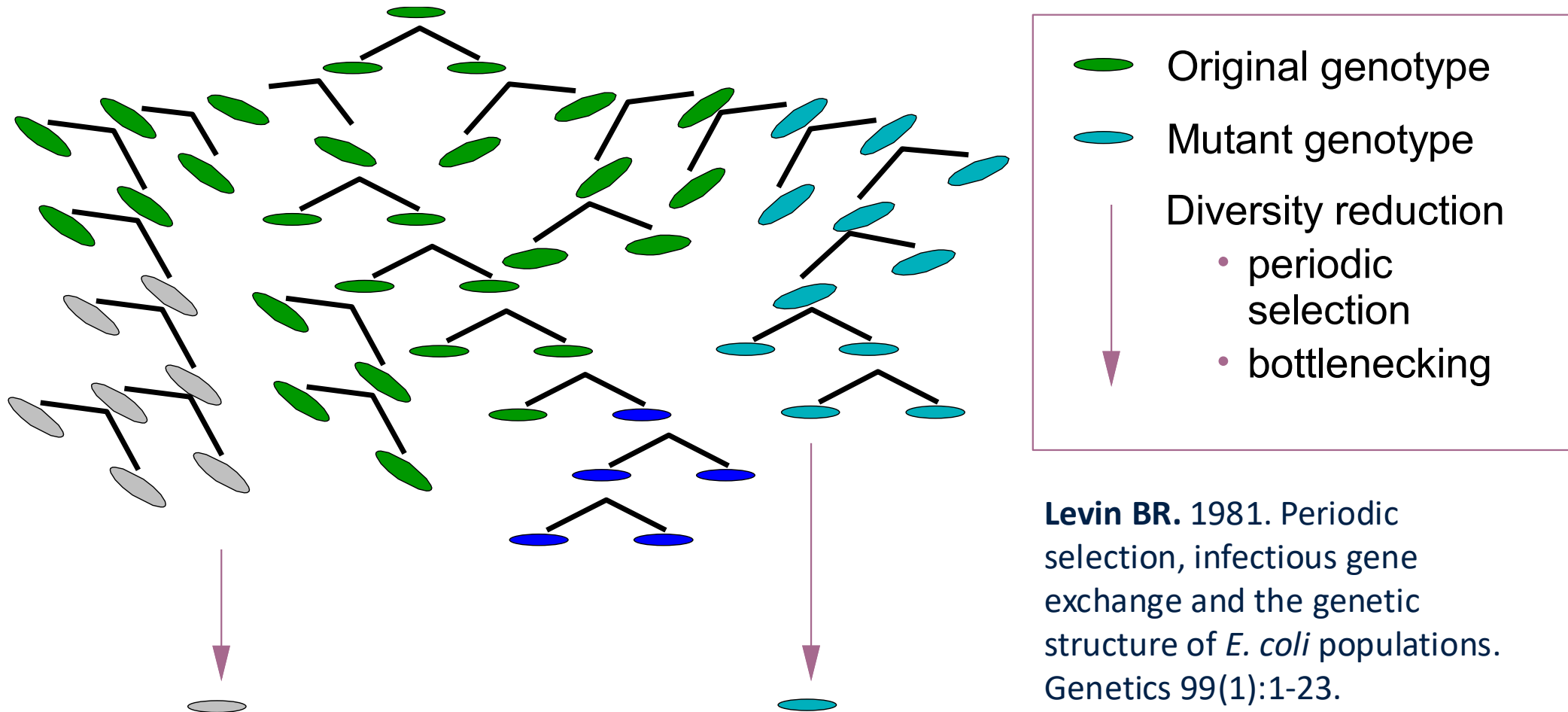
Ideas about bacterial populations are dominated
the facts that bacteria:

- are asexual;
- reproduce by binary fission, with each 'mother' cell giving rise to two identical 'daughter' cells (**clones**);
- accumulate genetic change by 'vertical' inheritance.



Gupta, S. & Maiden, M.C.J. (2001). Exploring the evolution of diversity in pathogen populations. *Trends in Microbiology* **9**, 181-192.

The clonal population model: asexuality with diversity reduction



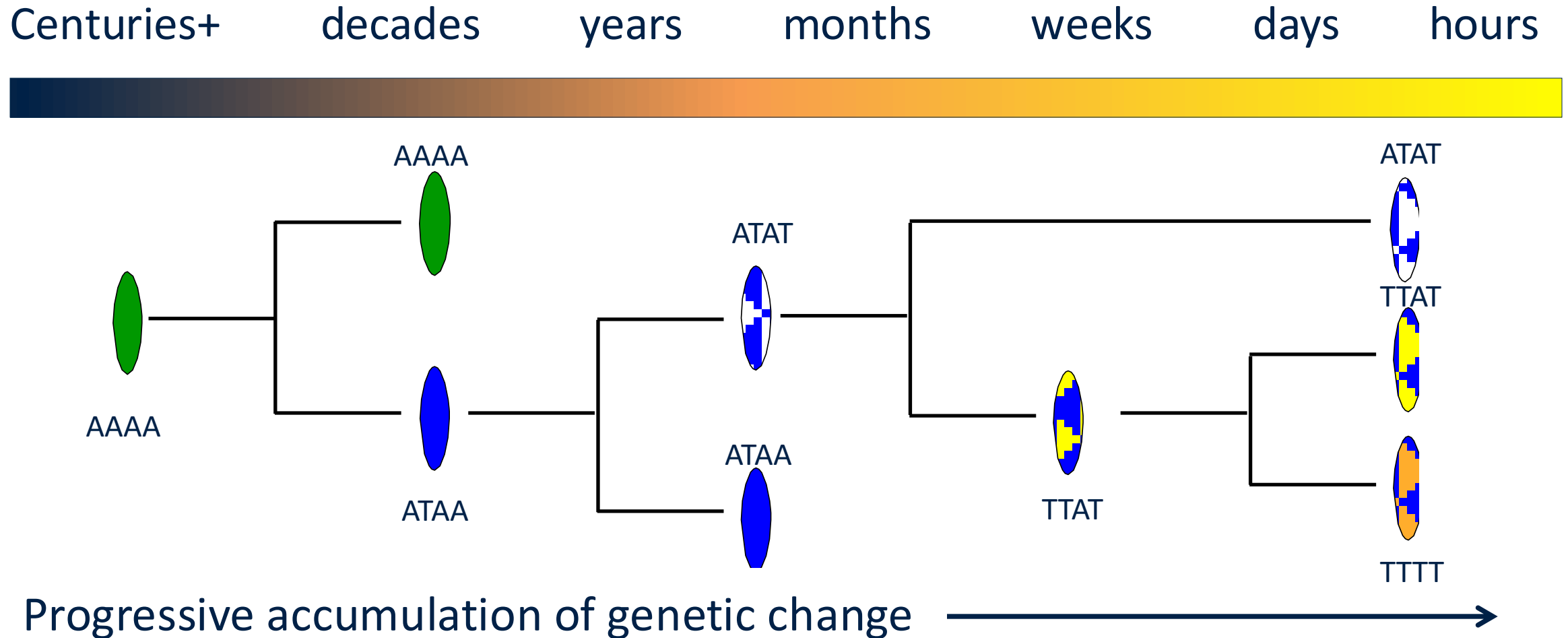
Levin BR. 1981. Periodic selection, infectious gene exchange and the genetic structure of *E. coli* populations. *Genetics* 99(1):1-23.

Within this model, Bacterial populations are (relatively) easy to understand.

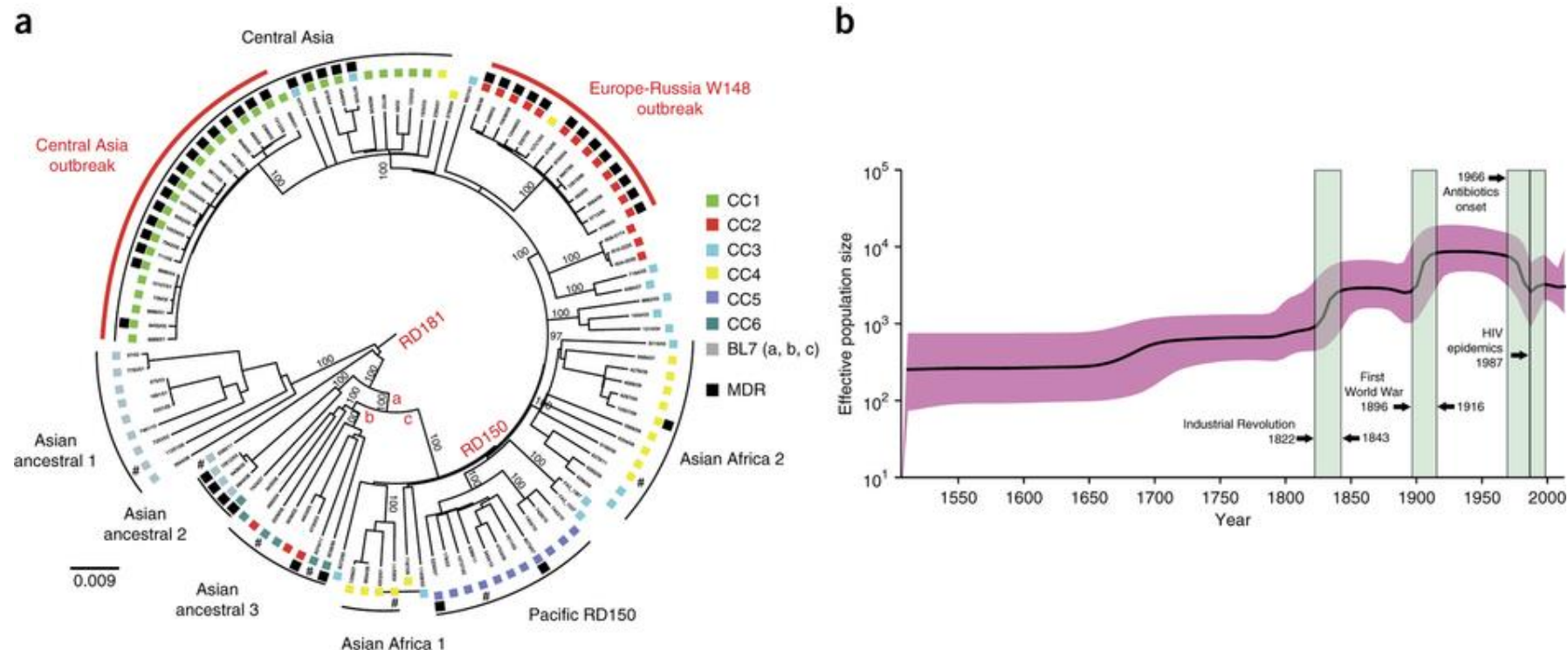
Selection pressures and impact on sequences

- **Diversifying (directional, positive) selection** – changes fitness phenotype
 - e.g. nonsynonymous mutations leading to antigenic diversity or antimicrobial resistance and mutations that change expression.
- **Stabilising selection (purifying, negative) selection** – retains phenotype and fitness
 - the gene or its expression is unchanged e.g. ‘housekeeping genes’.
- **Neutral variation** – variation that does not impact on fitness phenotype
 - synonymous changes may or may not be neutral in any gene.

Molecular epidemiology made easy: clonality

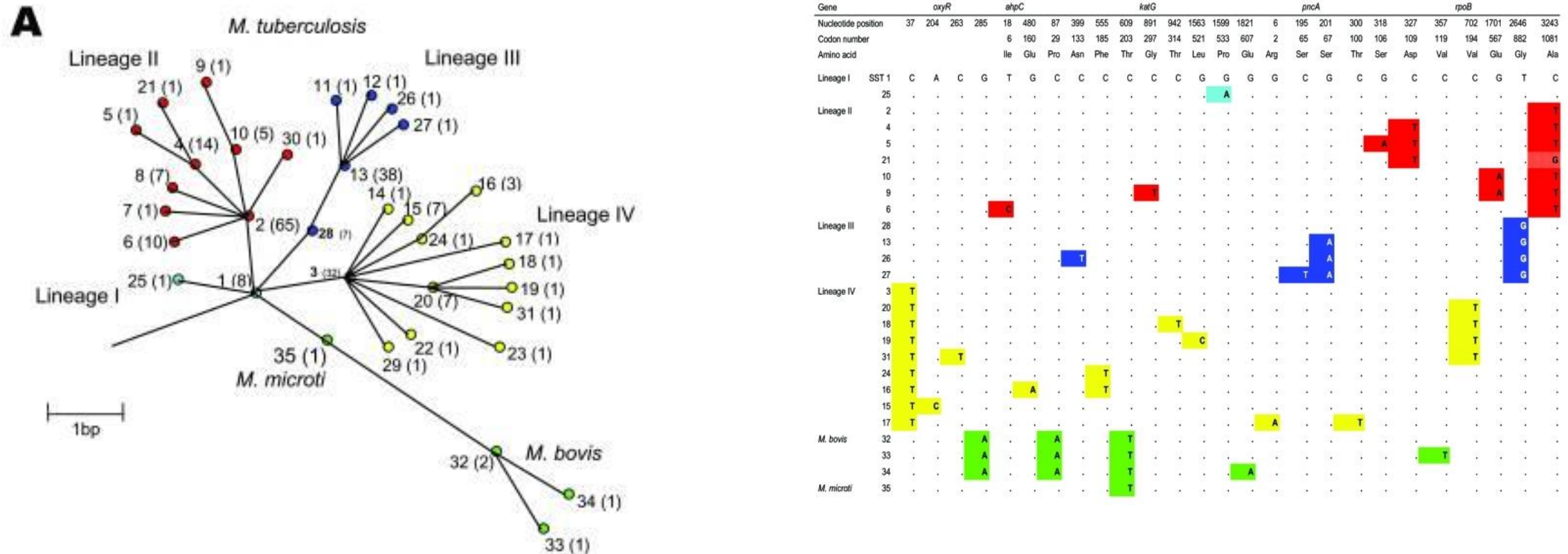


Mycobacterium tuberculosis is clonal: spread of the Beijing lineage



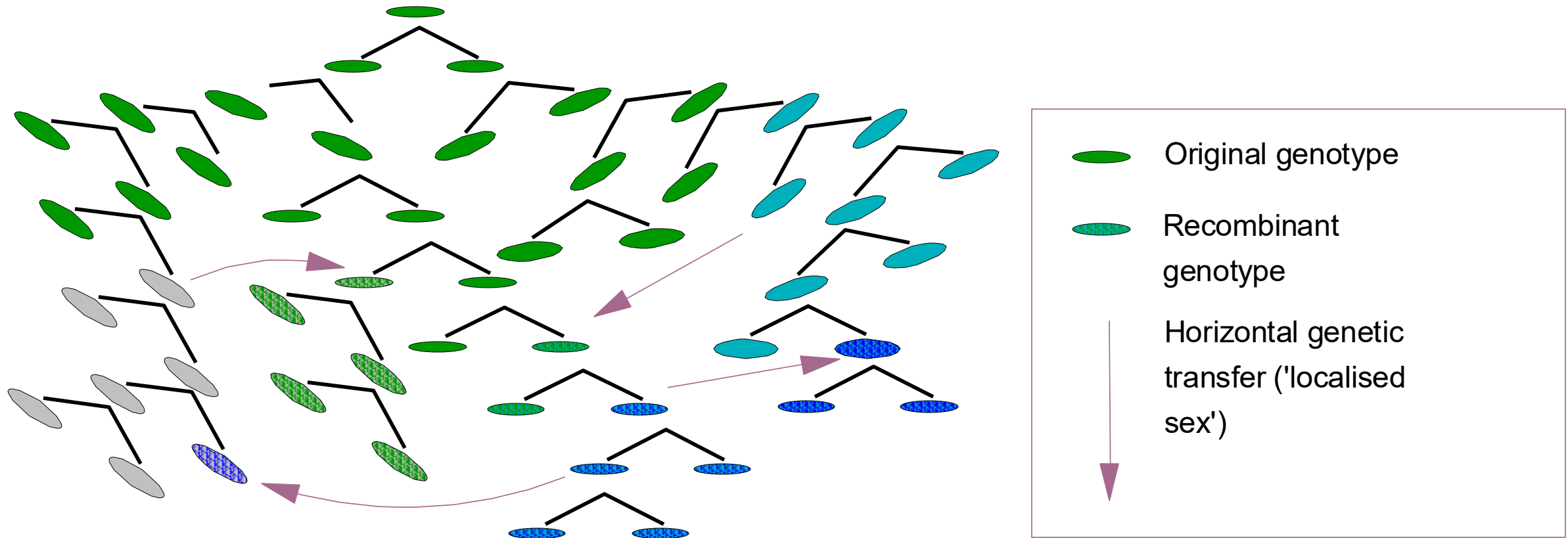
Merker, M., Blinet. Al. (2015). Evolutionary history and global spread of the *Mycobacterium tuberculosis* Beijing lineage. *Nat Genet* **47**, 242-249.

Mycobacterium tuberculosis lineages resolved by synonymous (silent) substitutions



Baker, L., Brown, T., Maiden, M. C. & Drobniewski, F. (2004). Silent nucleotide polymorphisms and a phylogeny for *Mycobacterium tuberculosis*. *Emerging Infectious Diseases*. **10**, 1568-1577.

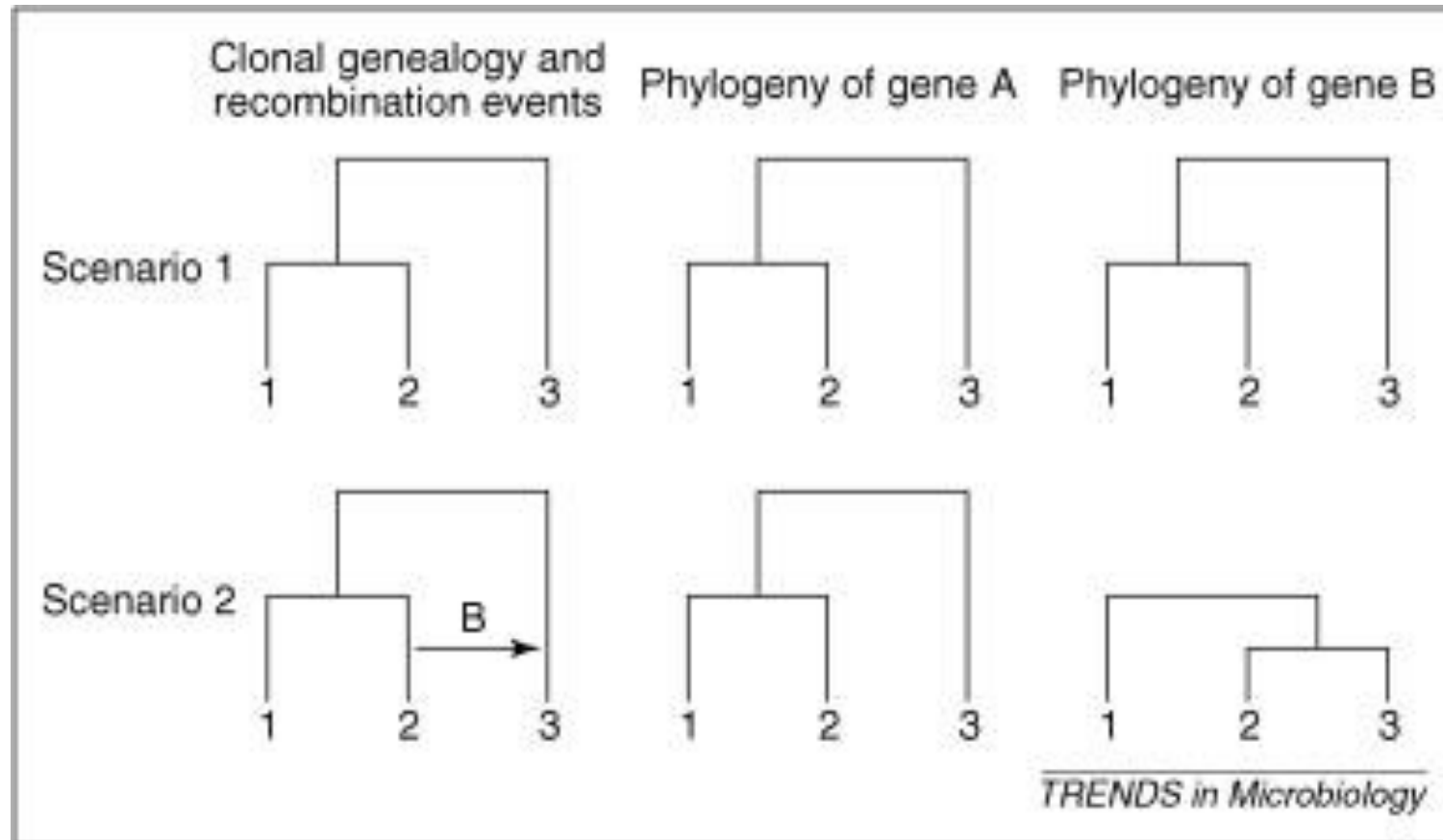
Impact of Horizontal genetic exchange (HGT) on bacterial population structure



HGT disrupts clonal structure, breaking down tree-like phylogeny, linkage disequilibrium and congruence.

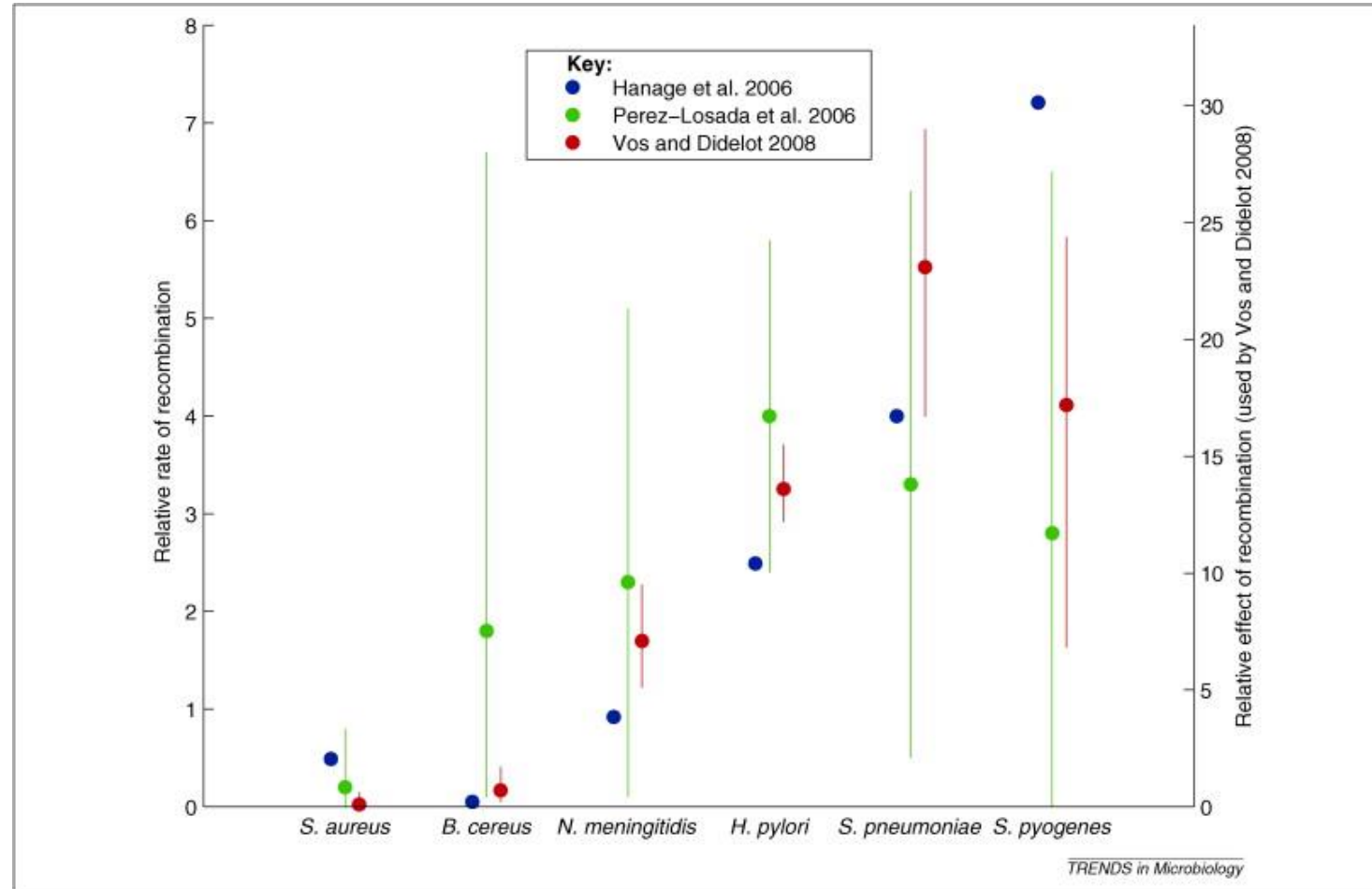
Maynard Smith J, Dowson CG, Spratt BG. (1991). Localized sex in bacteria. *Nature* 349, 29-31.

Recombination and phylogeny



Didelot, X. & Maiden, M. C. (2010). Impact of recombination on bacterial evolution. *Trends in Microbiology* **18**, 315-322.

Varying impact of recombination



Didelot, X. & Maiden, M. C. (2010). Impact of recombination on bacterial evolution. *Trends in Microbiology* 18, 315-322.

Clonal and non-clonal population structures

Clonal

- Linkage disequilibrium
 - non-random allele combinations.
- Tree-like phylogeny
 - a bifurcating tree accurately models descent.
- Congruence
 - the same phylogenetic signal is recorded throughout the genome.

Non-clonal

- Linkage equilibrium
 - random allele combinations.
- Net-like phylogeny
 - a bifurcating tree cannot model descent.
- Incongruence
 - different phylogenetic signals are recorded throughout the genome.

Maiden, M. C. (2006). Multilocus Sequence Typing of Bacteria. *Annual Review of Microbiology* 60, 561-588.

Dealing with partial HGT

HGT violates the assumptions of clonal evolution and can be accounted for in models of bacterial evolution by:

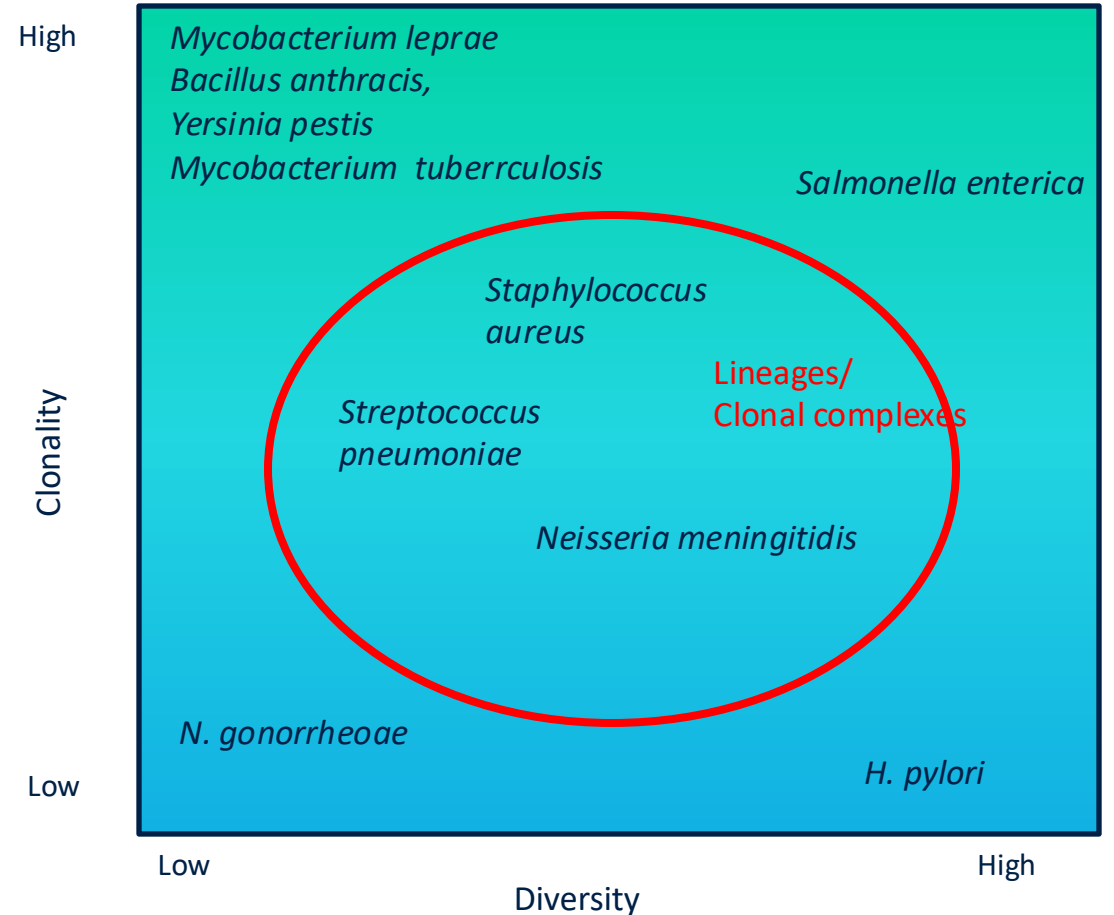
- Ignoring it, which only works if HGT is uncommon;
- Using loci and alleles as the unit of analysis (*e.g.* gene-by-gene MLST approaches);
- Identifying likely HGT and removing it (*e.g.* GUBBINS);
- Estimating relative contributions of recombination and mutation in evolutionary models (*e.g.* ClonalFrame ML).

Population structure and diversity

In conclusion:

- different levels of clonal signal are observed in different bacterial populations;
- this is a consequence of differing relative rates of recombination to mutation;
- however, other forces will also play a role.

Didelot, X. & Maiden, M. C. J. (2010). Impact of recombination on bacterial evolution. *Trends Microbiol* 18, 315-322.



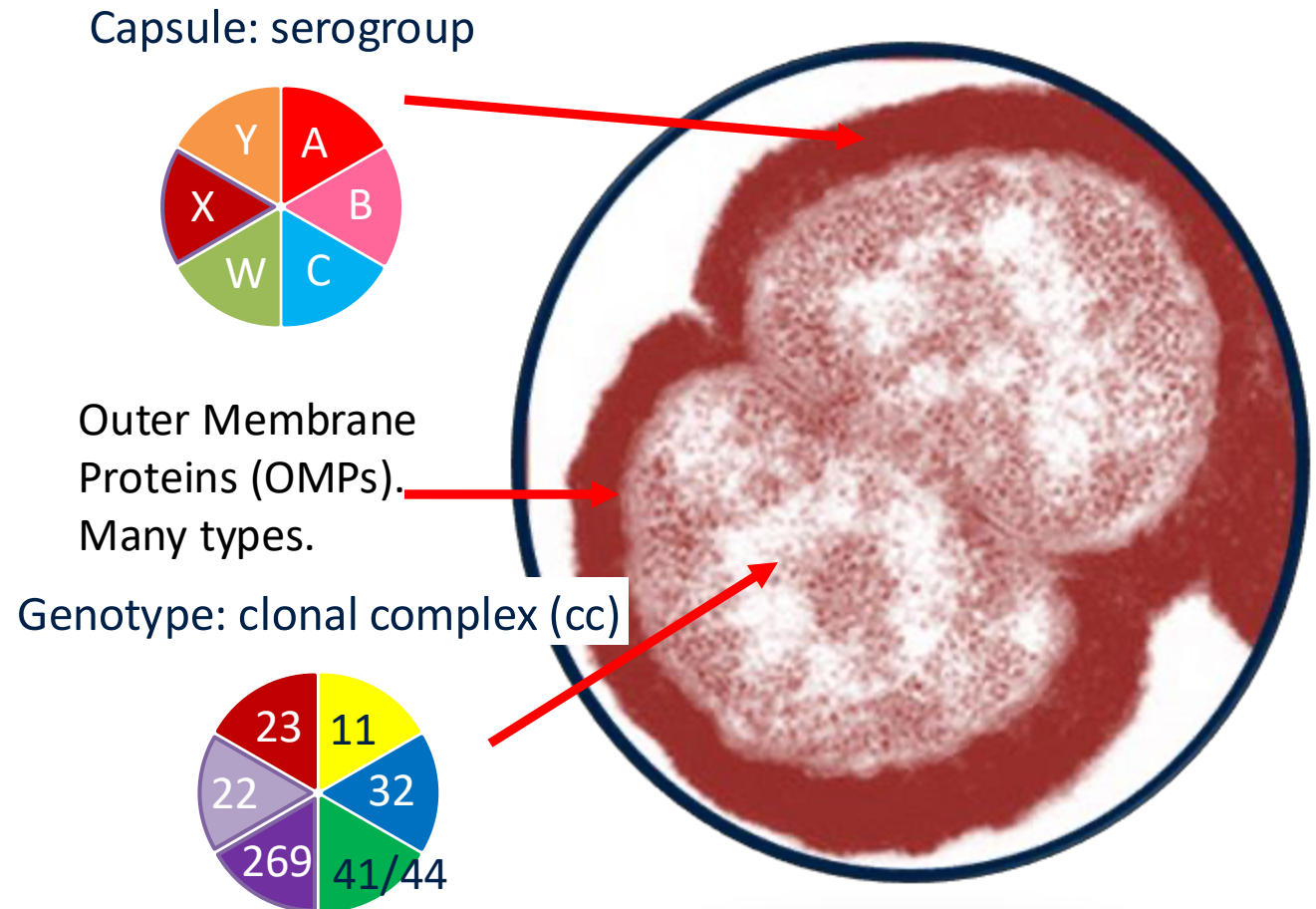
Summary

- Bacterial populations are extremely ancient,
 - bacterial diversity is almost unimaginably large.
- The bacterial population diversity has been shaped by selective forces,
 - positive (diversifying) selection, negative (stabilising) selection, neutrality.
- Extensive diversity is present within as well as among genomes,
 - HGT mobilises this diversity, as seen in the spread of AMR.
- Limitations in HGT enable species and distinct types to emerge,
 - the ratio of mutation to genetic transfer (recombination) affects population structure.

Within and among species diversity in the
Neisseria

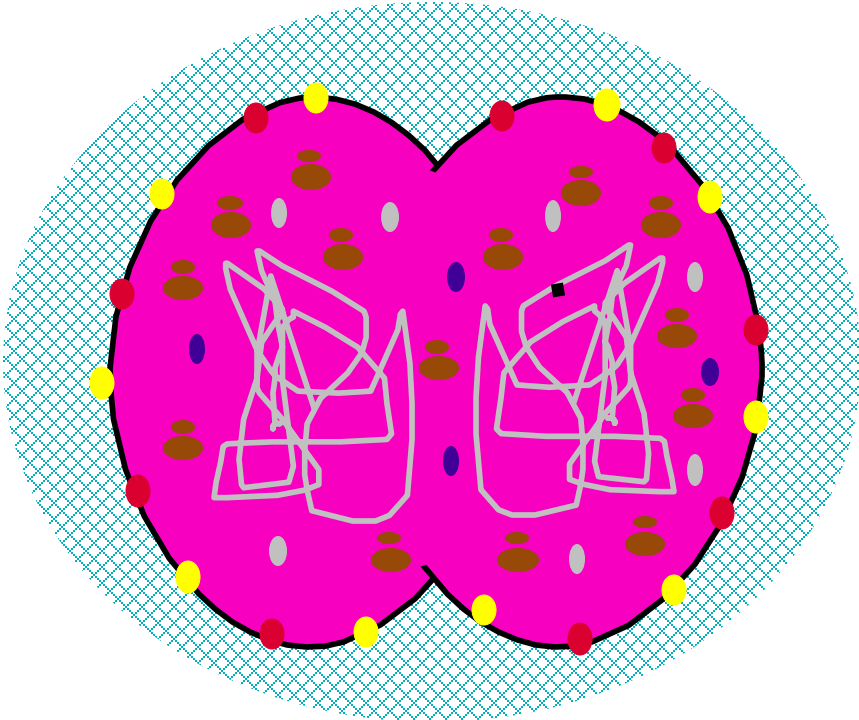
Within species diversity *Neisseria meningitidis*

- Meningococci are highly diverse antigenically and genetically,
 - this diversity is structured.
- 12 capsular serogroups,
 - 6 associated with invasive disease.
- Extensive evidence of HGT, but stable lineages are present,
 - these are associated with phenotypes, including invasive disease.



Ganesh, K., Allam, M., Wolter, N., Bratcher, H. B., Harrison, O. B., Lucidarme, J., Borrow, R., de Gouveia, L., Meiring, S., Birkhead, M., Maiden, M. C., von Gottberg, A. & du Plessis, M. (2017). Molecular characterization of invasive capsule null *Neisseria meningitidis* in South Africa. *BMC Microbiology* **17**, 40.

Neisseria meningitidis characterisation

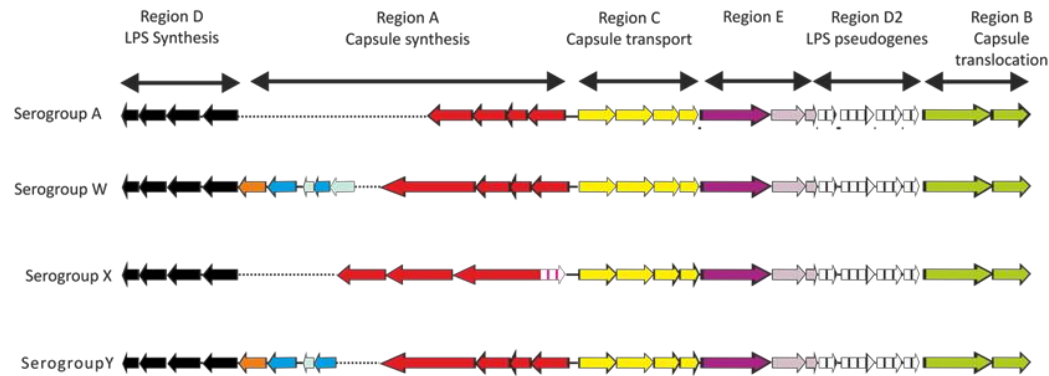


Component	Phenotypic	Genotypic
Capsule	Serogroup	<i>cps</i> region
OMPS	Serotype, Subtype, etc.	<i>porA</i> , <i>porB</i> , <i>fetA</i> , etc.
Housekeeping genes	MLEE	MLST
Ribosomes	MALDITOF	16s rRNA, rMLST

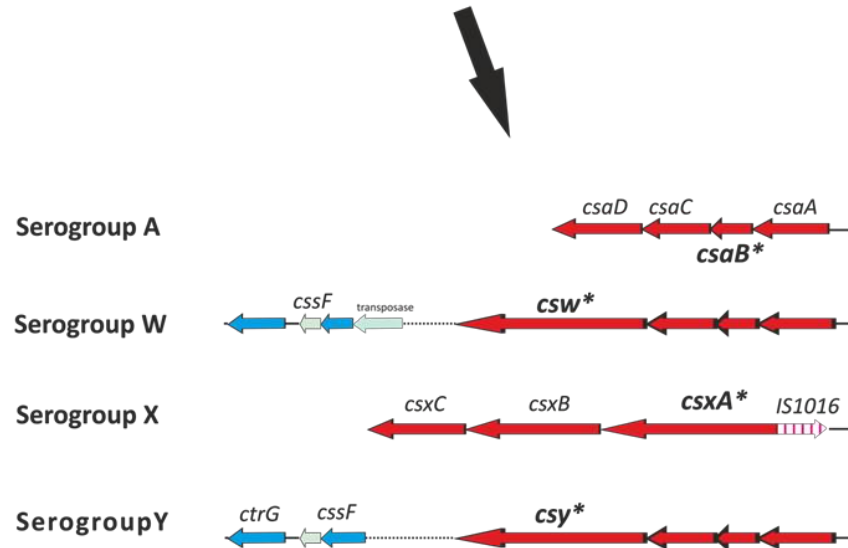
***Neisseria meningitidis* C: P1.21-15,16: F1-7: ST-10217 (cc10217)**

Jolley, K. A., Brehony, C. & Maiden, M. C. (2007). Molecular typing of meningococci: recommendations for target choice and nomenclature. *FEMS Microbiol Rev* **31**, 89-96.

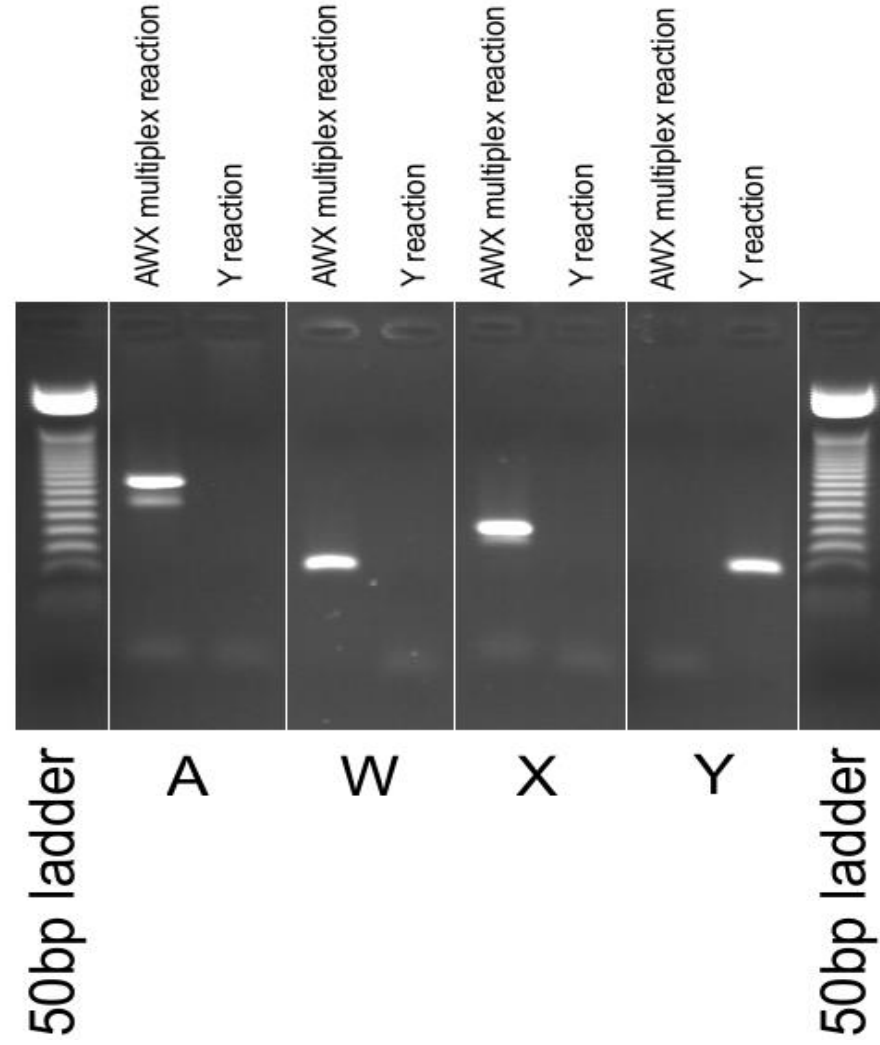
Meningococcal genogrouping: characterising the *cps* region encoding the capsule (serogroup antigen)



Harrison, O. B., Claus, H., Jiang, Y., Bennett, J. S., Bratcher, H. B., Jolley, K. A., Corton, C., Care, R., Poolman, J. T., Zollinger, W. D., Frasch, C. E., Stephens, D. S., Feavers, I., Frosch, M., Parkhill, J., Vogel, U., Quail, M. A., Bentley, S. D. & Maiden, M. C. J. (2013). Description and nomenclature of *Neisseria meningitidis* capsule locus. *Emerging Infectious Diseases* 19, 566-573.



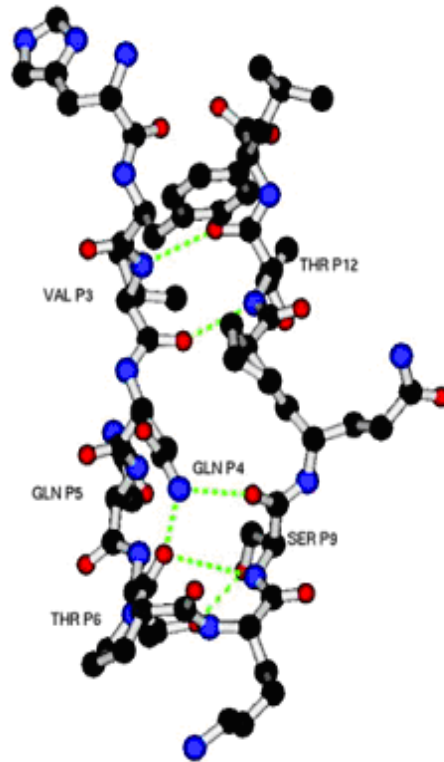
Genogrouping: a PCR will do it



PorA sequence variation and serosubtyping

- Major outer membrane protein (OMP);
- Serosubtyping antigen (originally with mAbs) and a major vaccine component;
- Most antigenic variability in two variable regions (VRs) of the protein and gene.
- A single PCR amplification and sequencing reaction can provide high discrimination among isolates.

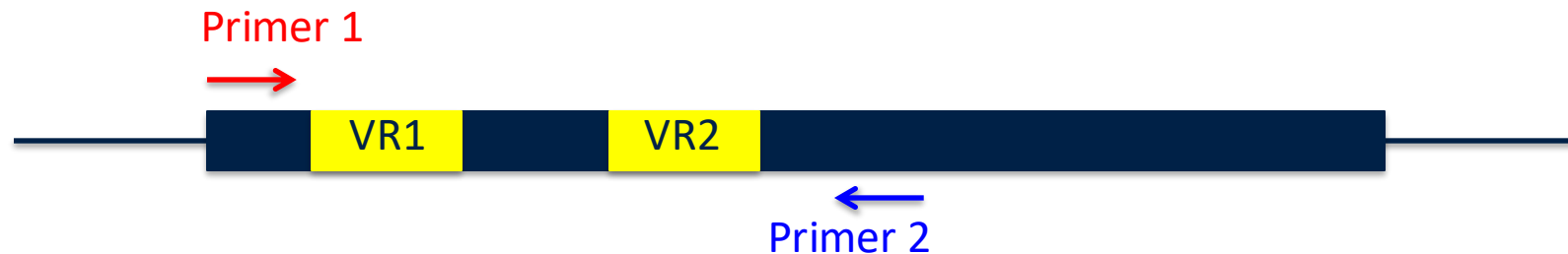
Structure of PorA P1.2 variant



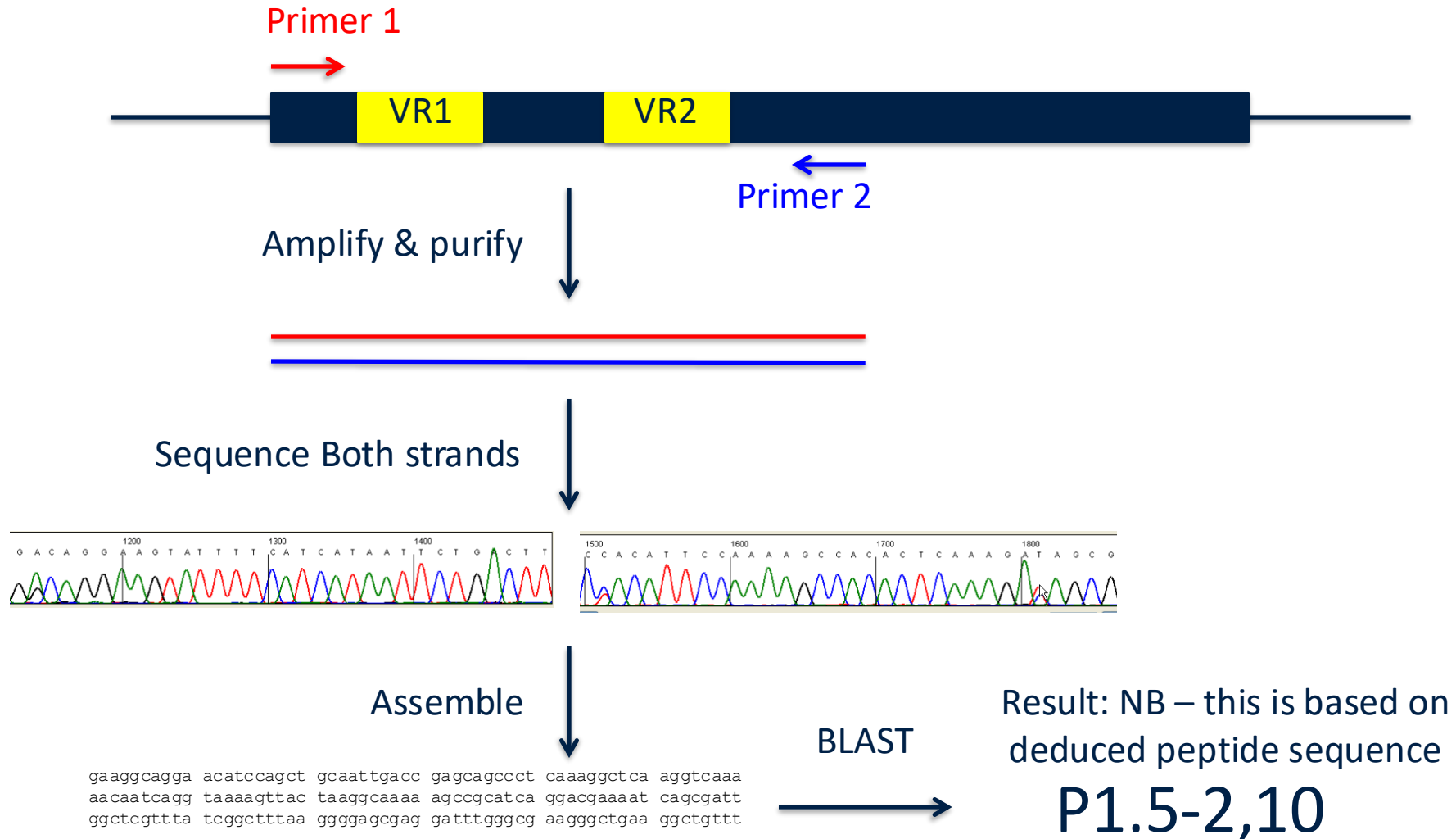
Tzitzilonis, C., Prince, S. M., Collins, R. F., Achtman, M., Feavers, I. M., Maiden, M. C. & Derrick, J. P. (2006).
Structural variation and immune recognition of the P1.2 subtype meningococcal antigen. *Proteins* 62, 947-955.

Sequence-based subtyping, PorA

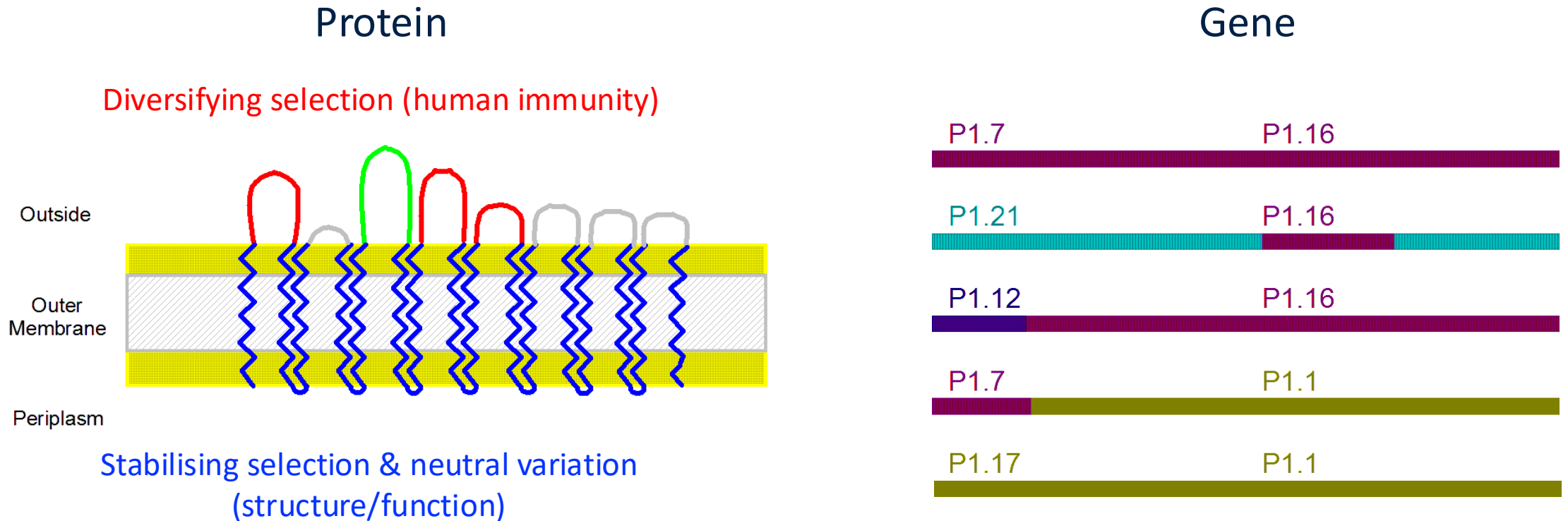
- The two most variable parts of the PorA protein (VR1 and VR2) are close together,
 - one sequencing reaction can determine the parts of the gene encoding both.
- *porA* gene fragment sequencing therefore extracts a lot of information for the amount of sequencing performed.



Sequencing *porA* VR1 and VR2



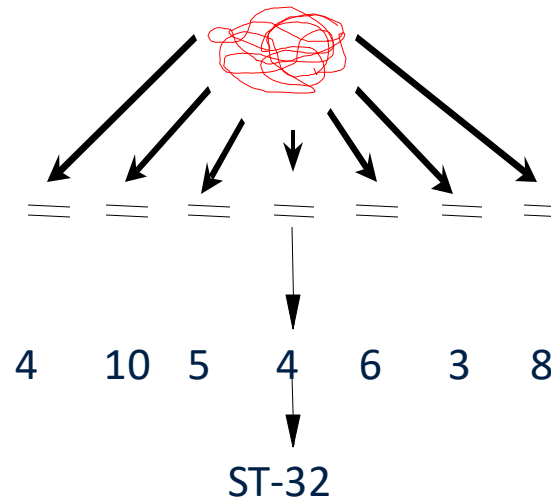
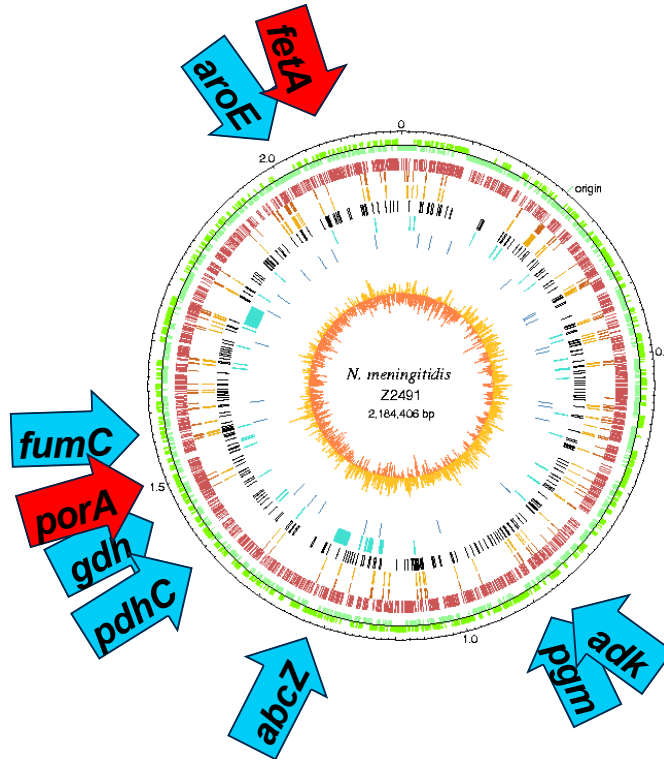
Meningococcal horizontal genetic transfer: mosaic *porA* genes



Maiden MCJ, Suker J, McKenna AJ, Bygraves JA, Feavers IM. Comparison of the class 1 outer membrane proteins of eight serological reference strains of *Neisseria meningitidis*. Mol. Microbiol. 1991;5:727-736.

Feavers IM, Heath AB, Bygraves JA, Maiden MC. Role of horizontal genetic exchange in the antigenic variation of the class 1 outer membrane protein of *Neisseria meningitidis*. Molecular Microbiology 1992;6:489-495.

Multi-locus typing in the meningococcus



DNA sample

PCR Amplicons

Curated sequences

Sequence type

B: P1.7,16: F5-1: ST-32 (cc32)

Antigen type
(fine type)

Sequence type &
clonal complex

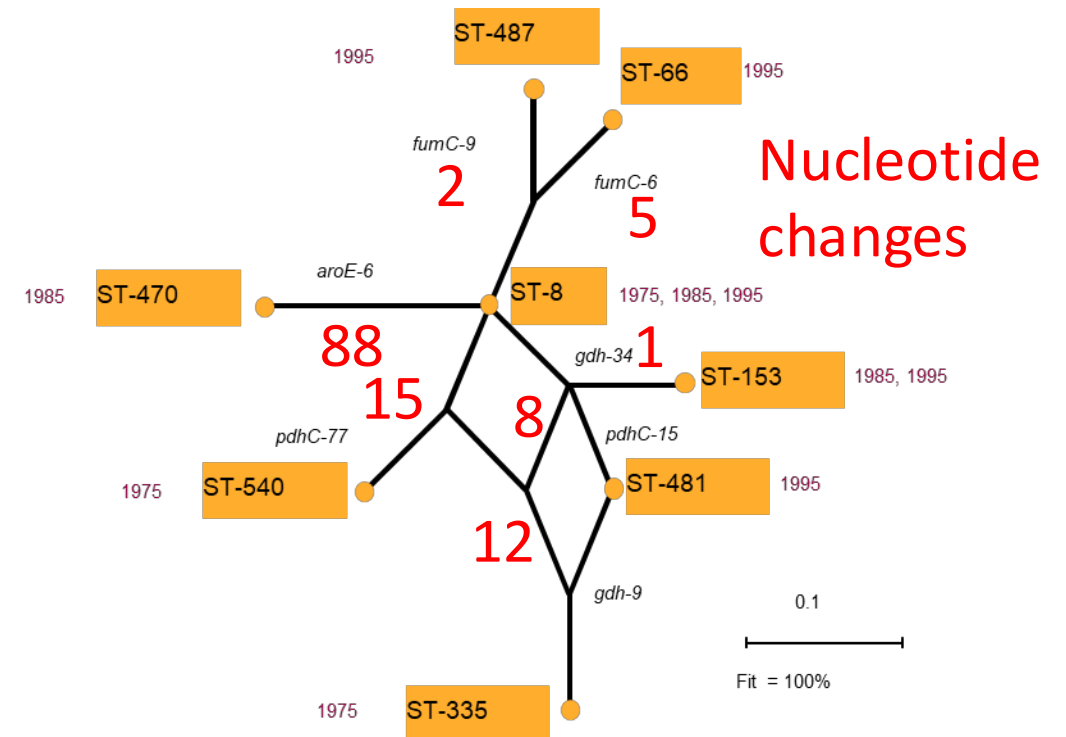
Maiden, MCJ, Bygraves, JA, Feil, E, Morelli, G, Russell, JE, Urwin, R, Zhang, Q, Zhou, J, Zurth, K, Caugant, DA, Feavers, IM, Achtman, M & Spratt, BG. 1998. Multilocus sequence typing: a portable approach to the identification of clones within populations of pathogenic microorganisms. *Proc Natl Acad Sci USA* 95, 3140-3145.

Maiden, MC. 2006. Multilocus Sequence Typing of Bacteria. *Annu Rev Microbiol* 60, 561-588.

Jolley KA, Brehony C, Maiden MC. 2007. Molecular typing of meningococci: recommendations for target choice and nomenclature. *FEMS Microbiol Rev* 31, 89-96.

MLST: allele-based analyses, sequence types (STs) and clonal complexes (ccs)

ST	<i>adk</i>	<i>abcZ</i>	<i>aroE</i>	<i>fumC</i>	<i>gdh</i>	<i>pdhC</i>	<i>pgm</i>
8	2	3	7	2	8	5	2
66	2	3	7	6	8	5	2
153	2	3	7	2	34	5	2
335	2	3	7	2	9	15	2
470	2	3	6	2	8	5	2
481	2	3	7	2	9	5	2
487	2	3	7	9	8	5	2
540	2	3	7	2	8	77	2



ST-8 Clonal Complex: cc8

Russell, J. E., Urwin, R., Gray, S. J., Fox, A. J., Feavers, I. M. & Maiden, M. C. (2008). Molecular epidemiology of meningococcal disease in England and Wales 1975-1995, before the introduction of serogroup C conjugate vaccines. *Microbiology* 154, 1170-1177.

Exercise: comparison of gene-by-gene with nucleotide-by-nucleotide analysis

Gene-by-gene (allele)

- <https://microreact.org/project/cMLFuQRKUJSFhXF2BGxQf8-bigsdb3598604244496374733861>
- <https://tinyurl.com/yvva8ud8>

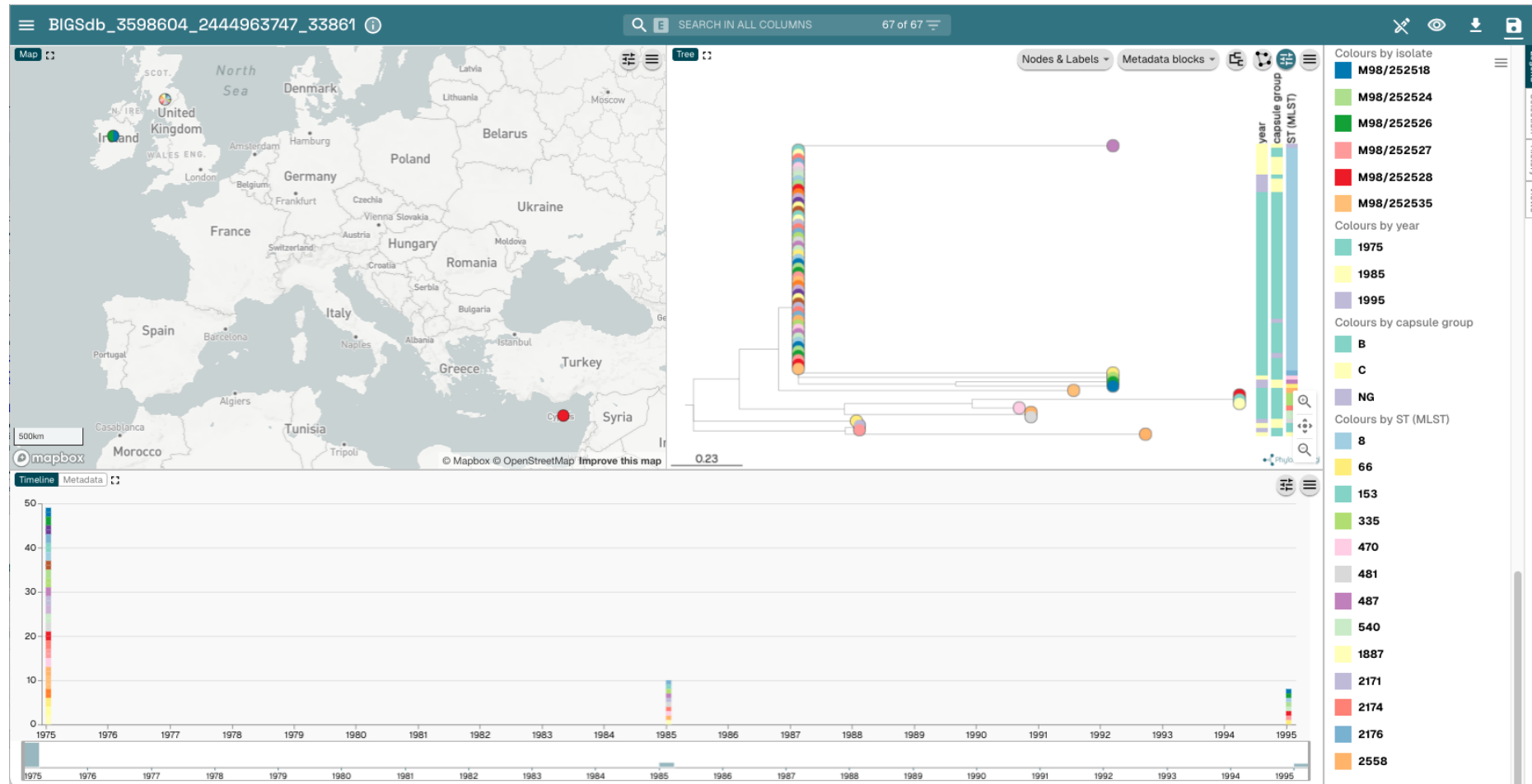
Nucleotide-by-nucleotide

- <https://microreact.org/project/nTP7f8chBnqo7S4pDyMNVV-bigsdb832730304944249432499>
- <https://tinyurl.com/2sb2axpv>

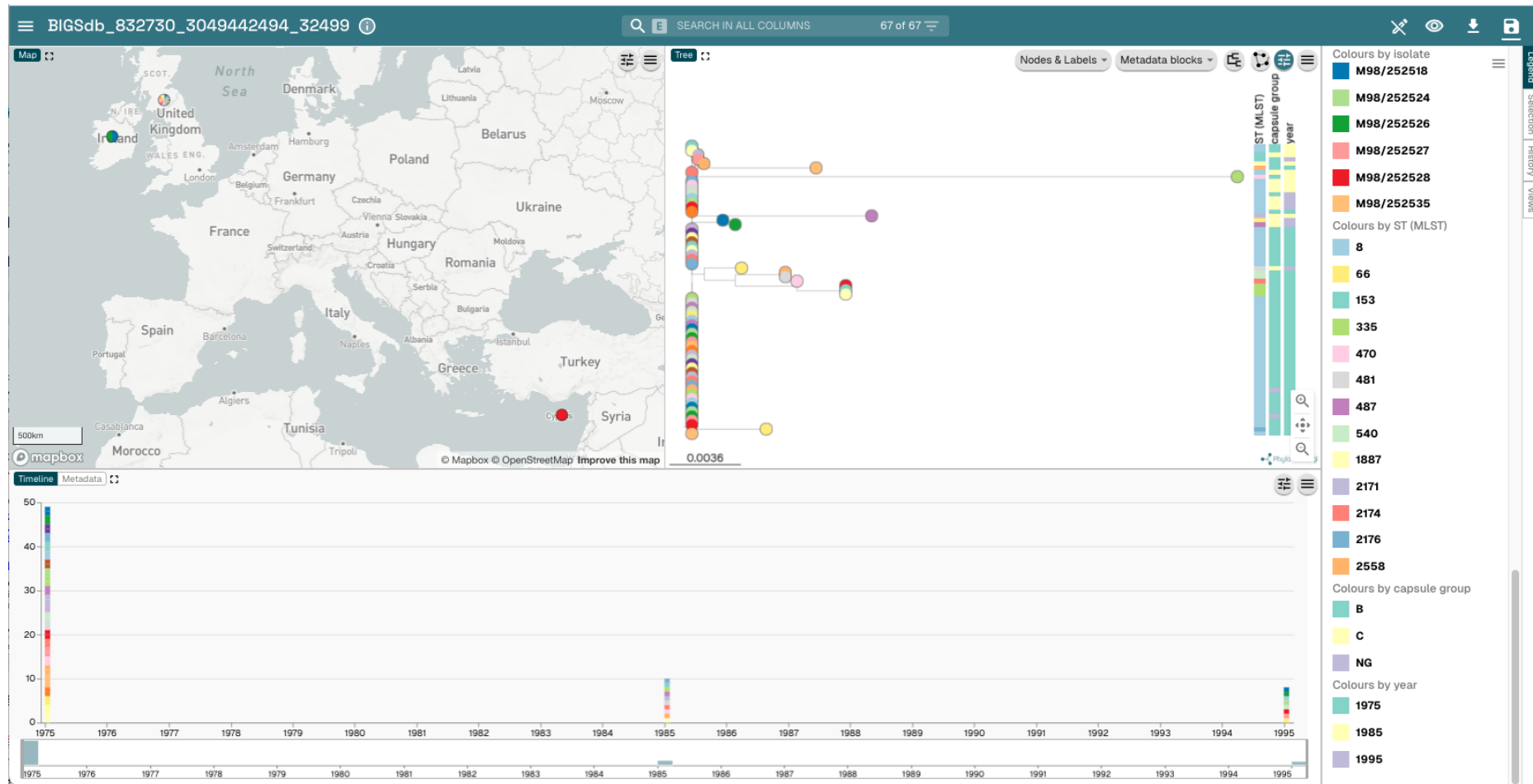
The trees are calculated from seven locus MLST data for 67 cc8 IMD isolates obtained in the UK in 1975, 1985, and 1995. Gene-by-gene analysis compensates for the impact of HGT, which distorts tree analyses by introducing multiple nucleotide changes in single genetic events .

Russell, J. E., Urwin, R., Gray, S. J., Fox, A. J., Feavers, I. M. & Maiden, M. C. (2008). Molecular epidemiology of meningococcal disease in England and Wales 1975-1995, before the introduction of serogroup C conjugate vaccines. *Microbiology* 154, 1170-1177.

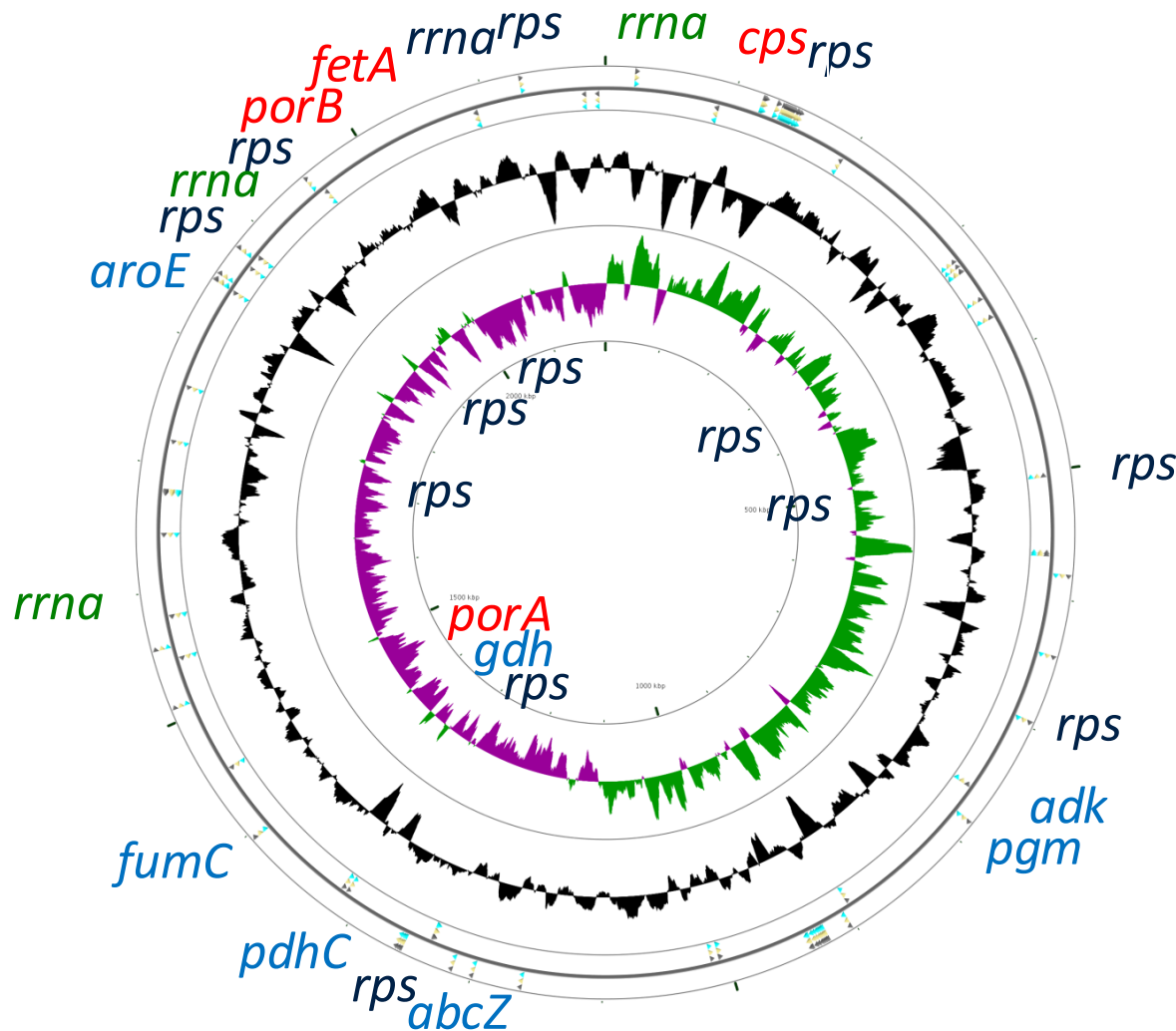
Gene-by-gene



Nucleotide-by-nucleotide



Typing targets in the meningococcal genome



Antigen genes:

- Capsule (*cps*), serogroup;
- OMPs (*porA*, *porB*, *fetA*), serotype, subtype fine type;
- BAST antigens (Bexsero vaccine).

Housekeeping genes:

- Ribosomal RNA genes (*rrna*) genus identification;
- Ribosome protein genes (*rps*), species and sub species characterisation (rMLST);
- MLST (metabolism), subspecies identification
- cgMLST, high-resolution typing.

Antibiotic Resistance genes.

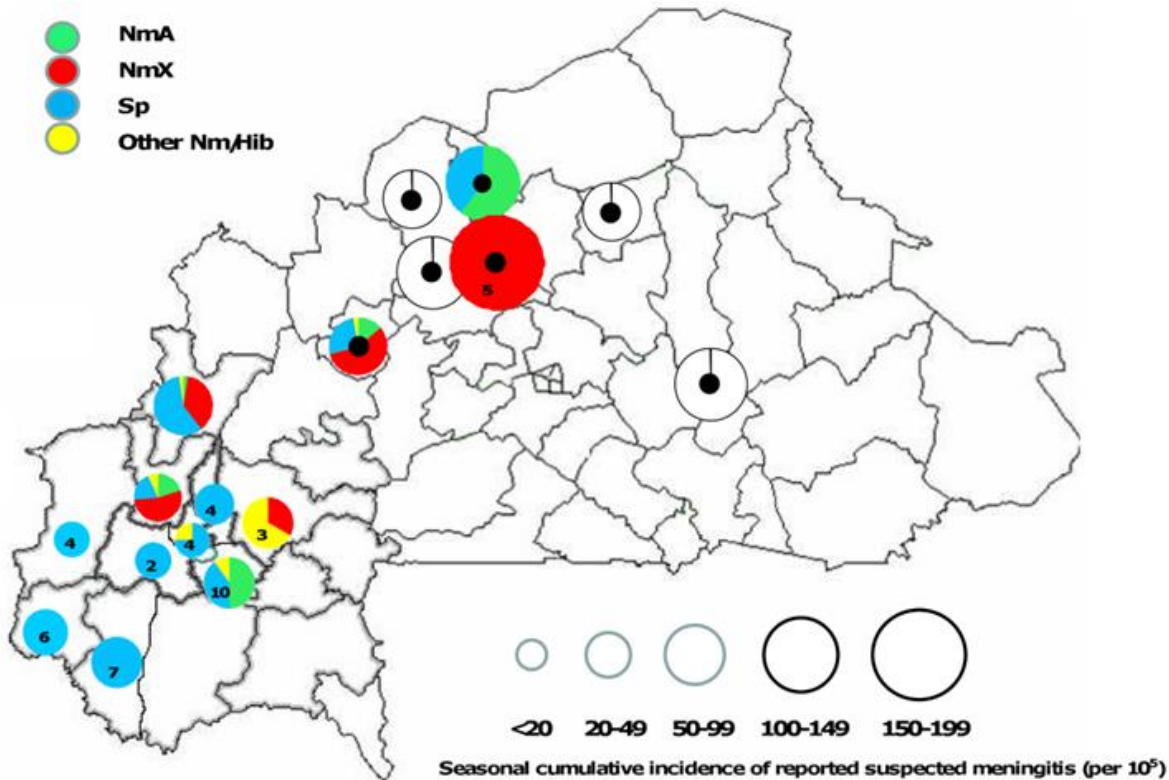
Practical 3: Introduction to molecular sequence typing using PubMLST typing

Serogroup C, W, and X meningococcal disease occurs in Africa

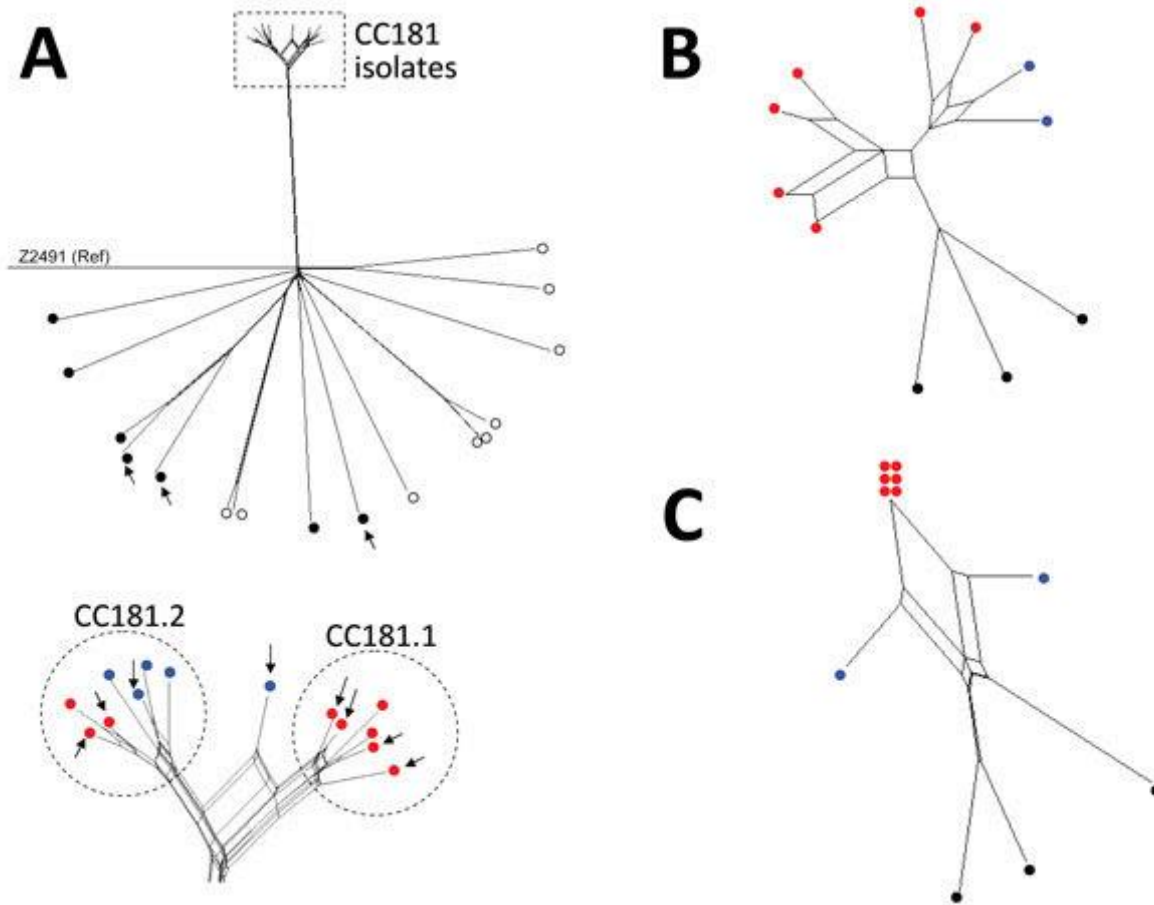
Wang, J. F., Caugant, D. A., Morelli, G., Koumaré, B. & Achtman, M. (1993). Antigenic and epidemiological properties of the ET-37 complex of *Neisseria meningitidis*. *Journal of Infectious Diseases* 167, 1320-1329.

Kwara, A., Adegbola, R. A., Corrah, P. T., Weber, M., Achtman, M., Morelli, G., Caugant, D. A. & Greenwood, B. M. (1998). Meningitis caused by a serogroup W135 clone of the ET-37 complex of *Neisseria meningitidis* in West Africa. *Trop Med Int Health* 3, 742-746.

Delrieu, I., Yaro, S., Tamekloe, T. A., Njanpop-Lafourcade, B. M., Tall, H., Jaillard, P., Ouedraogo, M. S., Badziklou, K., Sanou, O., Drabo, A., Gessner, B. D., Kambou, J. L. & Mueller, J. E. (2011). Emergence of epidemic *Neisseria meningitidis* serogroup X meningitis in Togo and Burkina Faso. *PLoS One* 6, e19513.



Meningococcal Serogroup X in sub-Saharan Africa



Key:

- White circles indicate carriage isolates;
- Black circles indicate invasive isolates obtained outside of Africa;
- Red circles indicate isolates obtained in sub-Saharan Africa since 2006;
- Blue circles indicate isolates obtained from sub-Saharan Africa during the 1990s.

Agmememel, A., Hong, E., Giorgini, D., Nunez-Samudio, V., Deghmane, A. E. & Taha, M. K. (2016). *Neisseria meningitidis* Serogroup X in Sub-Saharan Africa. *Emerg Infect Dis.* **22**, 698-702.

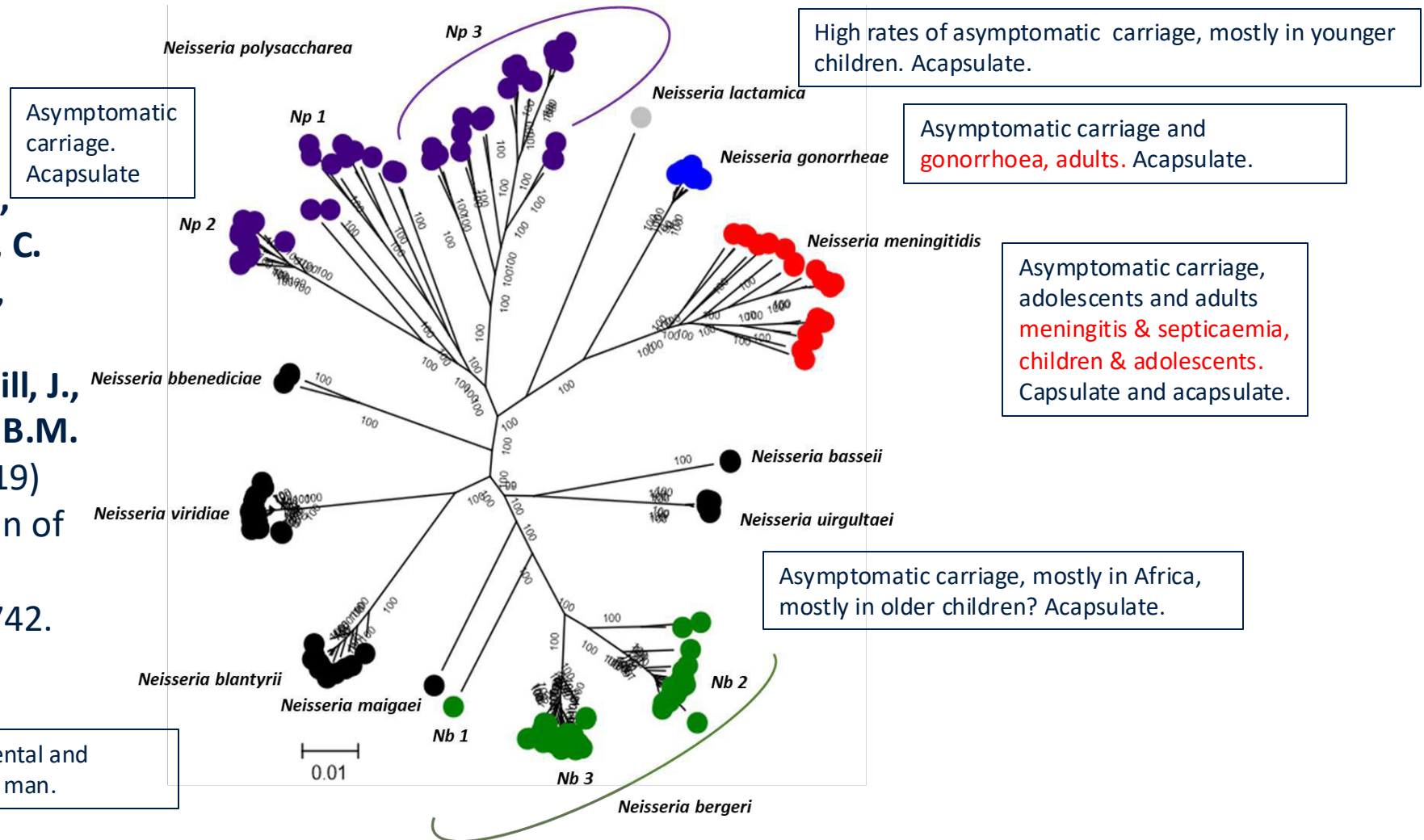
What is a bacterial species?

- In animals, and particularly mammals, species are defined by the ability to produce fertile offspring,
 - this does not, however, work well for other organisms.
- The concept of bacterial species is surprisingly controversial,
 - in terms of whether they exist, what they are, and how they are defined.
- From a clinical point of view, pragmatic definitions are important,
 - e.g. *Yersina pestis* and *Bacillus anthracis* are both genetically very close to their relatives, yet sufficiently clinically important and biologically distinct to warrant species designations.
- Pragmatically, bacterial species comprise organisms that have distinct phenotypes (e.g. causing plague or anthrax) and genotypes,
 - usually they are maintained by limitations of genetic exchange with close relatives.

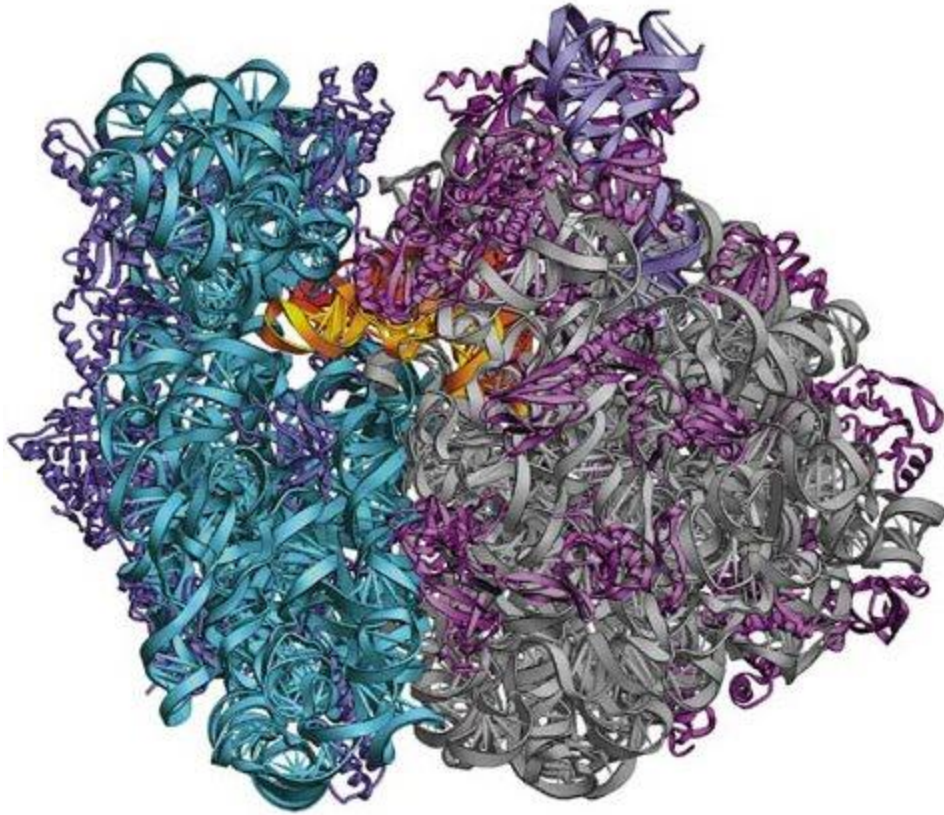
What is a species? Genus *Neisseria*, cgMLST 95% ANI threshold

Diallo, K., MacLennan, J., Harrison, O.B., Msefula, C., Sow, S.O., Daugla, D.M., Johnson, E., Trotter, C., MacLennan, C.A., Parkhill, J., Borrow, R., Greenwood, B.M. and Maiden, M.C.J. (2019) Genomic characterization of novel *Neisseria* species *Scientific Reports*, **9**, 13742.

Other species are associated with dental and mucosal colonisation in animals and man.



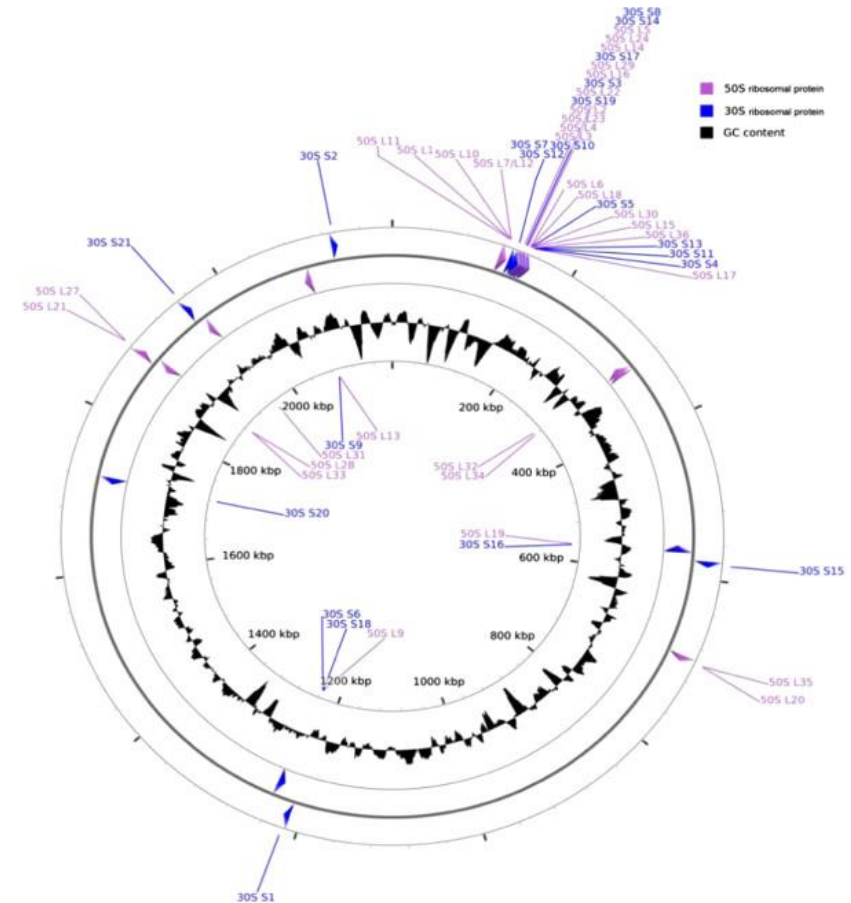
Ribosomal protein genes



- Present throughout bacterial domain.
- 53 genes – high resolution.
- Distributed throughout the genome.
- Encode complex protein structure - coevolution of genes.

Ribosomal MLST (rMLST)

- Ribosomal MLST provides a universal characterisation scheme
 - ‘from domain to strain’.
- Most bacteria have 53 protein subunits,
 - variation of the *rps* genes are used just as for conventional MLST loci.
 - These loci are least affected by HGT.
- This provides highly accurate species identification,
 - but does rely on WGS data.



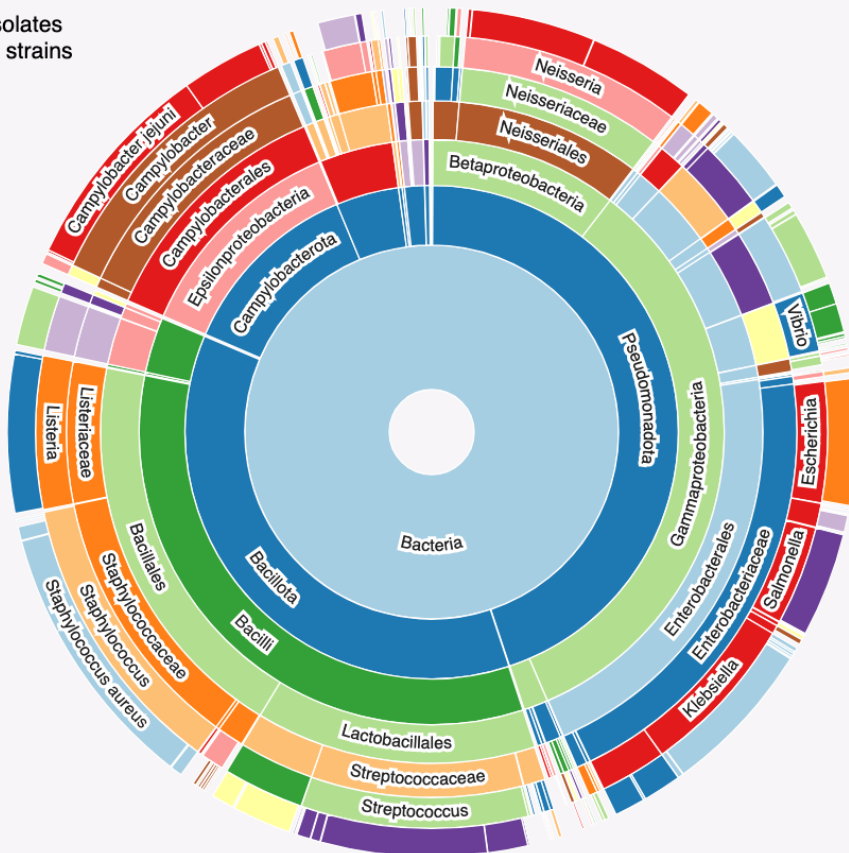
Jolley, K. A., Bliss, C. M., Bennett, J. S., Bratcher, H. B., Brehony, C. M., Colles, F. M., Wimalarathna, H. M., Harrison, O. B., Sheppard, S. K., Cody, A. J. & Maiden, M. C. (2012). Ribosomal Multi-Locus Sequence Typing: universal characterization of bacteria from domain to strain. *Microbiology*. **158**, 1005-1015.

Ribosomal MLST

Q IDENTIFY SPECIES

Genome taxonomic coverage

- ☒ all isolates
☐ type strains



Click on any arc to zoom in. Click on centre circle to zoom out. Click on background to reset zoom.

'Type strains' include exemplar records where no formal type strain has been defined.

Ribosomal Multilocus Sequence Typing (rMLST) is an approach that indexes variation of the 53 genes encoding the bacterial ribosome protein subunits (*rps* genes) as a means of integrating microbial taxonomy and typing.

The *rps* gene variation catalogued in this database permits rapid identification of the phylogenetic position of any bacterial sequence at the domain, phylum, class, order, family, genus, species and strain levels.

rMLST is described in [Jolley *et al.* 2012 *Microbiology* 158:1005-15](#) .

Databases curated by [James Bray](#).

For academic non commercial use only. For full terms and conditions please see [rMLST_licence.pdf](#). To discuss any other sort of use, including a commercial use licence, please contact innovation@innovation.ox.ac.uk quoting reference 7895/MC.

Access to the databases can be obtained by [registering for a PubMLST account](#) and making the request via the registration system.

- [Taxonomic coverage of the databases](#)
- [Programmatic access to the species identification tool](#)
- [Upload your own genome data privately to a scratch database for analysis](#)
- [Description of species identification method](#)



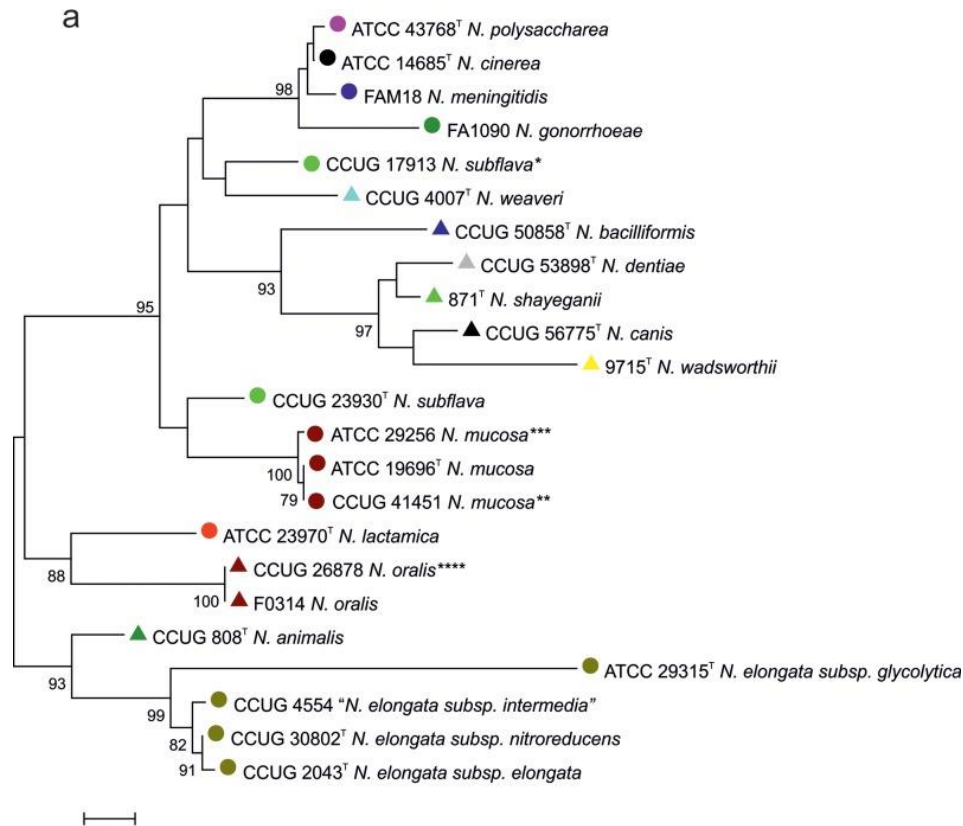
Species identification, the *rplF* assay

- **Problem:** rapid cost effective determination of *Neisseria* species from 1000s of isolates obtained in the MenAfriCar surveys.
 - 16s rRNA sequencing too cumbersome and insufficient discrimination or resolution.
- **Solution:** identify a short sequenceable gene fragment (~400bp in length) diagnostic for species.
- **Implementation:** Phylogeny of *Neisseria* species generated and compared to phylogenies of individual genes
 - A fragment of *rplF* gene was congruent with clusters in the phylogeny of all genes and used for the assay.

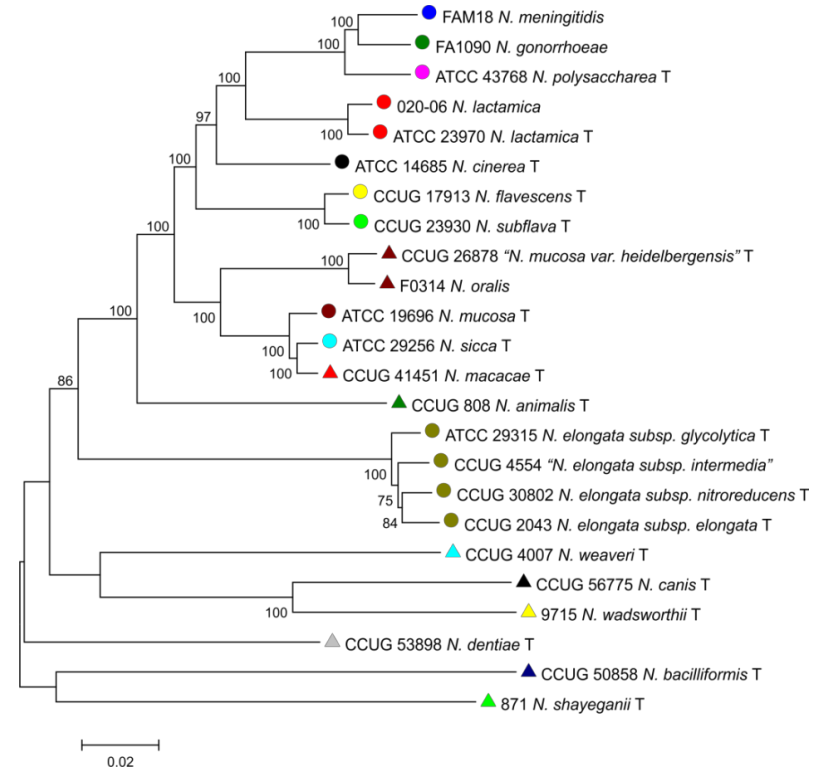
Bennett, J. S., Watkins, E. R., Jolley, K. A., Harrison, O. B. & Maiden, M. C. (2014). Identifying *Neisseria* species using the 50S ribosomal protein L6 (*rplF*) gene. *J Clin Microbiol* **52**, 1375-1381.

Comparison of 16s rRNA sequencing and *rplF* sequencing

16s rRNA gene fragment phylogeny (single rRNA encoding gene)



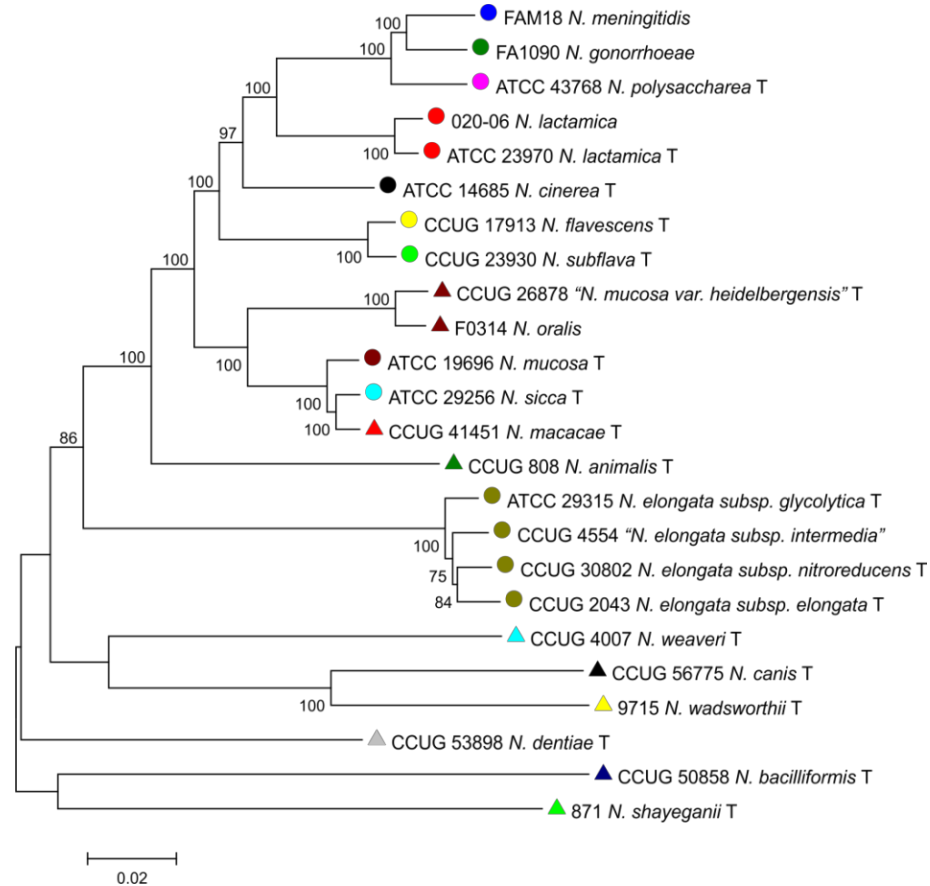
rMLST phylogeny (53 ribosomal protein encoding genes)



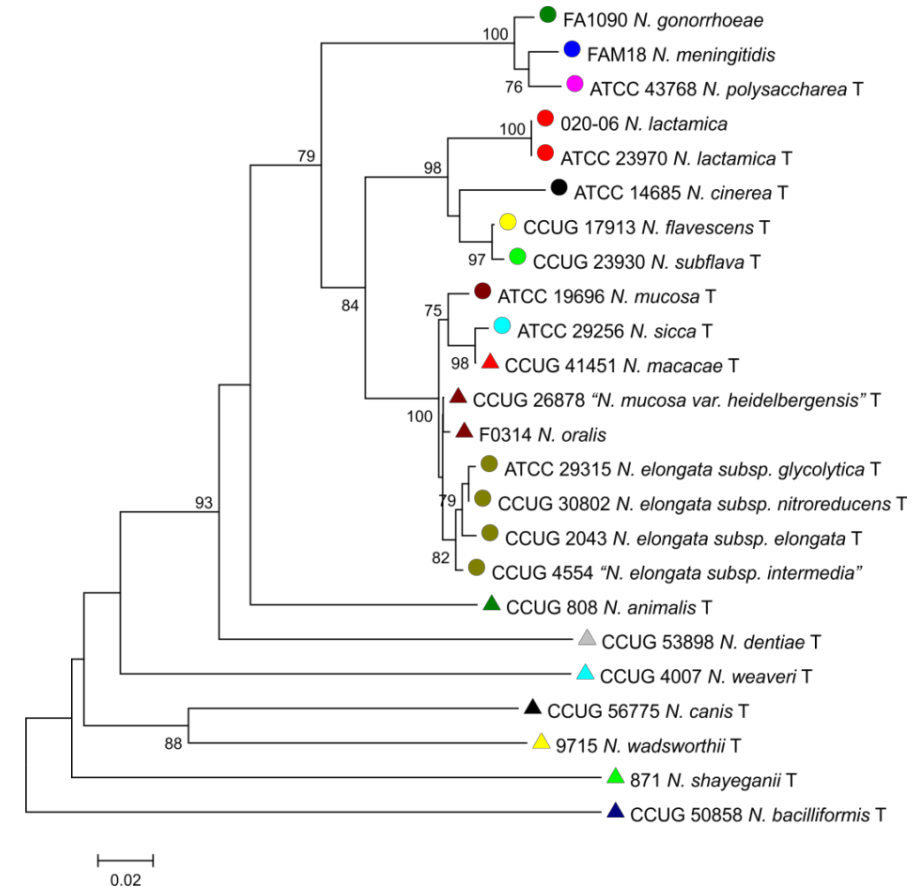
Bennett, J. S., Watkins, E. R., Jolley, K. A., Harrison, O. B. & Maiden, M. C. (2014). Identifying *Neisseria* species using the 50S ribosomal protein L6 (*rplF*) gene. *J Clin Microbiol* 52, 1375-1381.

Rapid species assignment: *rplF* sequence

rMLST phylogeny (53 genes)

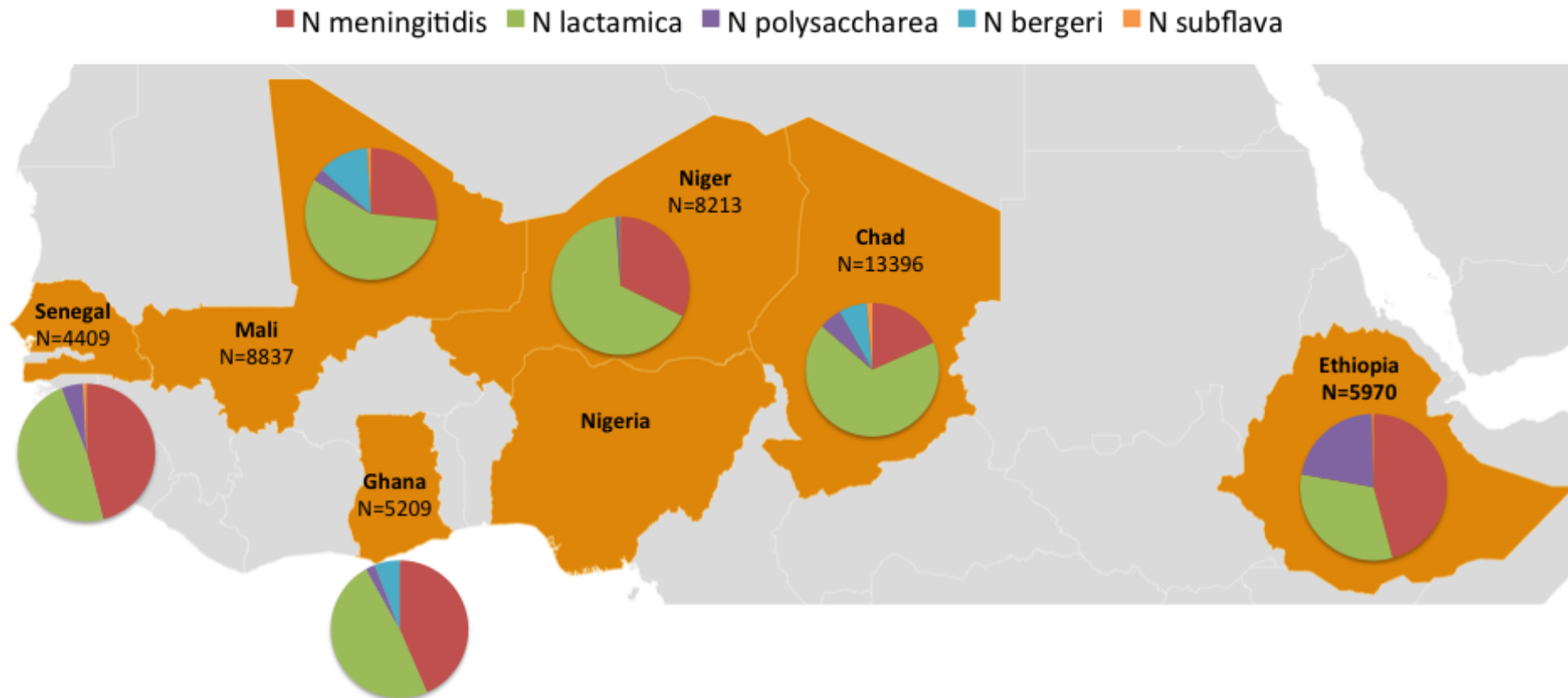


rplF fragment (413bp) phylogeny



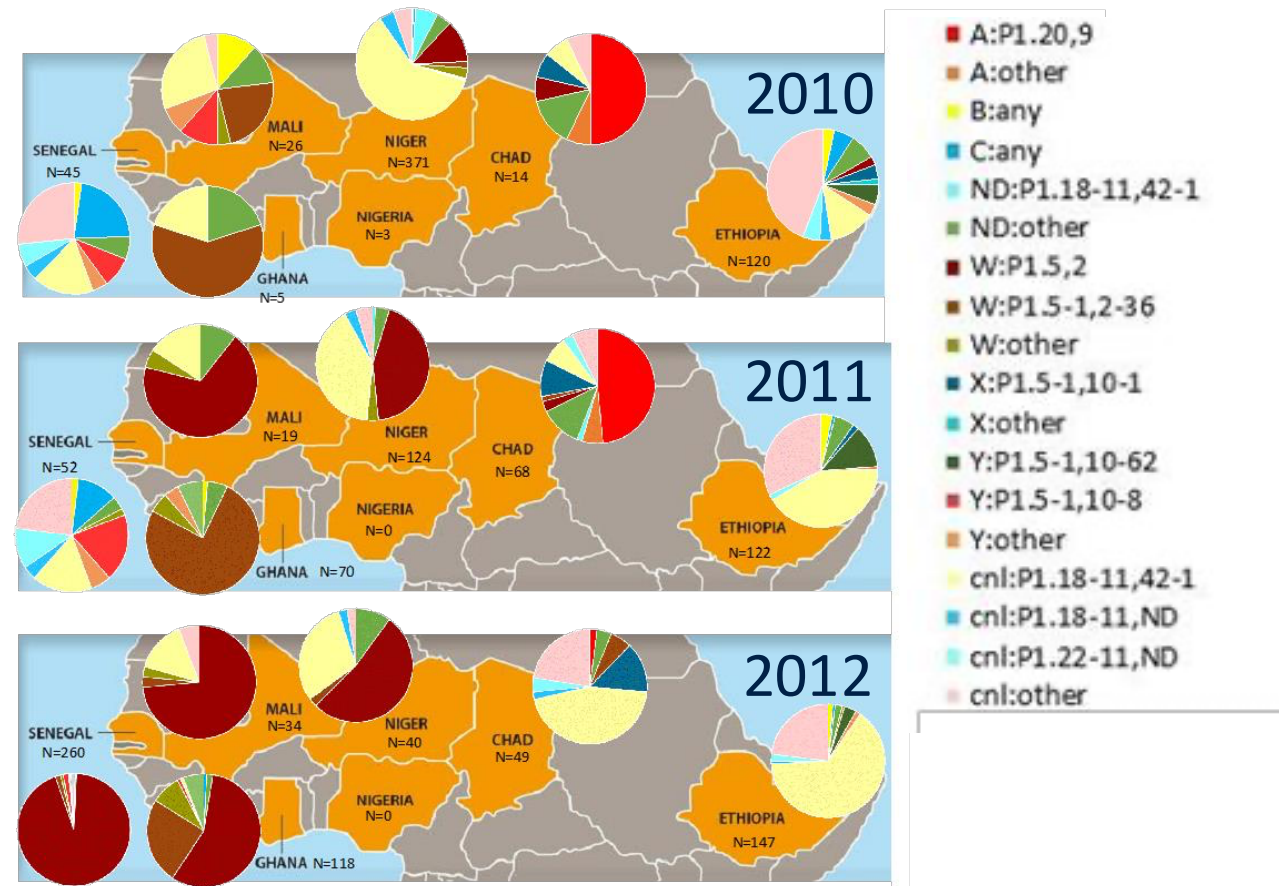
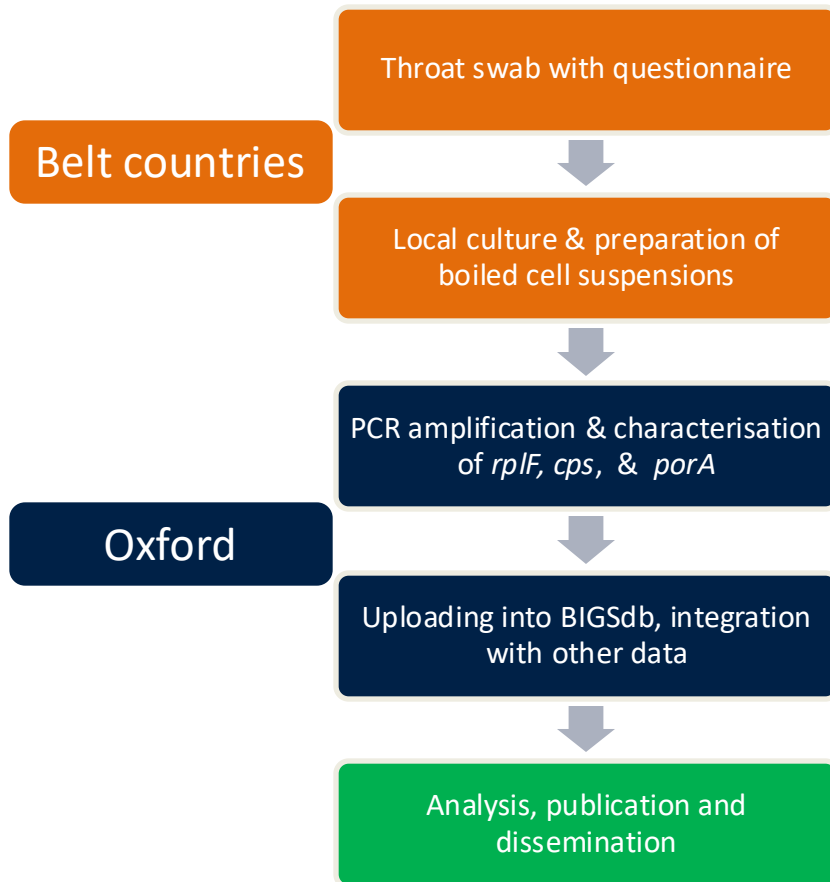
Bennett, J. S., Watkins, E. R., Jolley, K. A., Harrison, O. B. & Maiden, M. C. (2014). Identifying *Neisseria* species using the 50S ribosomal protein L6 (*rplF*) gene. *J Clin Microbiol* 52, 1375-1381.

Neisseria species distribution



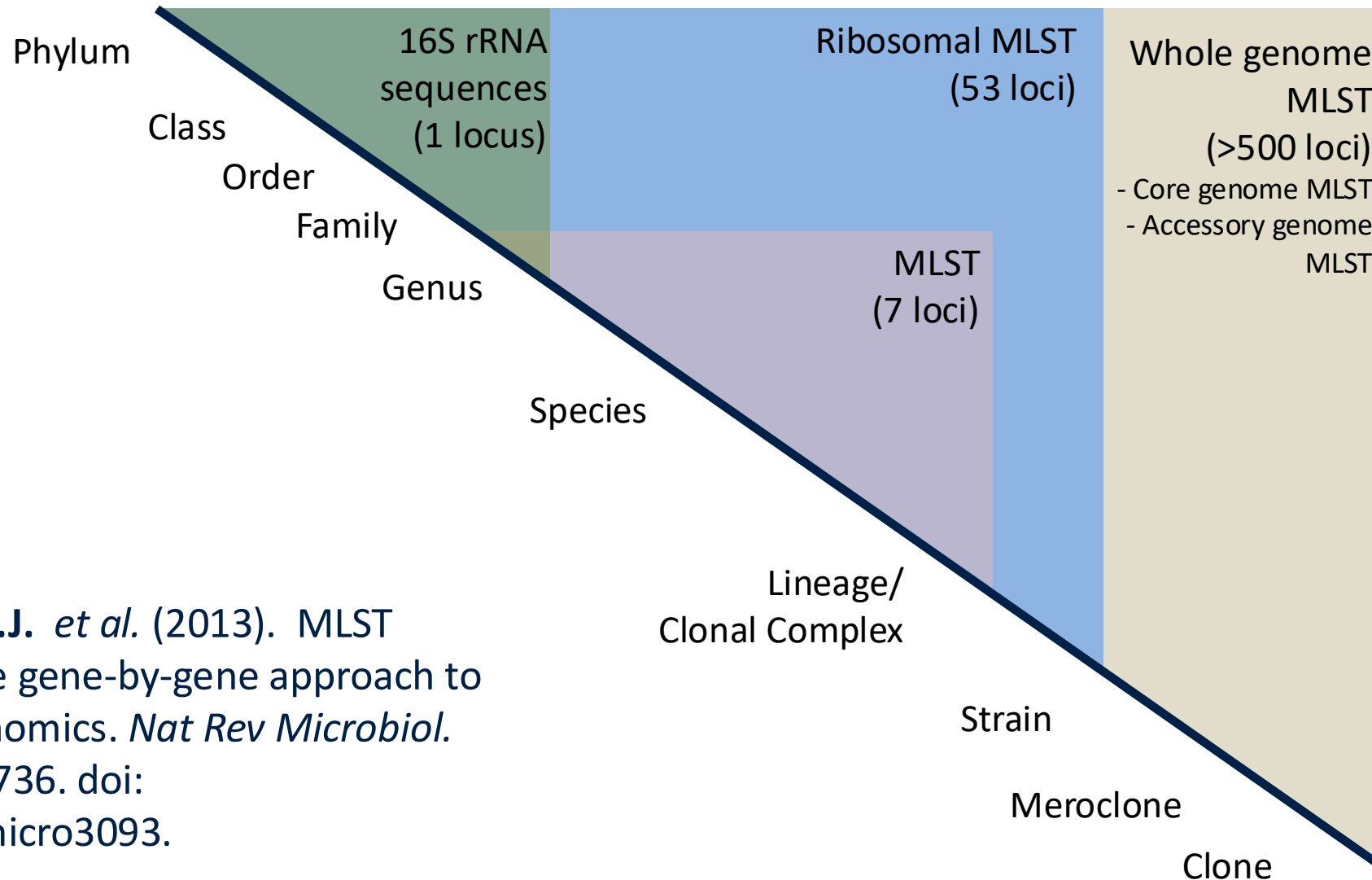
Diallo, K., et al., Greenwood, B. M. & Maiden, M. C. (2016). Pharyngeal carriage of *Neisseria* species in the African meningitis belt. *J Infect* **72**, 667-677.

MenAfriCar: meningococcal type distribution in carriage



Ali, O., et al. and MenAfriCar Consortium. (2015) The Diversity of Meningococcal Carriage Across the African Meningitis Belt and the Impact of Vaccination With a Group A Meningococcal Conjugate Vaccine. *J Infect Dis* **212**, 1298-1307

Sequence data and nomenclature



Maiden M.C.J. et al. (2013). MLST revisited: the gene-by-gene approach to bacterial genomics. *Nat Rev Microbiol.* 11(10), 728-736. doi: 10.1038/nrmicro3093.